

XSFN-Net: A Hybrid Deep Spiking Fractional Neural Network for OSCC Classification in Histopathological Images

*A Project Report submitted in the partial fulfillment of
the Requirements for the award of the degree*

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project that is entitled with the name **“XSFN-NET: A HYBRID DEEP SPIKING FRACTIONAL NEURAL NETWORK FOR OSCC CLASSIFICATION IN HISTOPATHOLOGICAL IMAGES”** is a bonafide work done by **U.Pavitra (22471A0564), Ch.Vijaya (22471A0512), Ch.Mounika (22471A0510), A.Arthi (22471A0501)** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in the Department of **COMPUTER SCIENCE AND ENGINEERING** during 2025-2026.

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PO11: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Project Course Outcomes (CO'S):

CO421.1: Analyse the System of Examinations and identify the problem.

CO421.2: Identify and classify the requirements.

CO421.3: Review the Related Literature

CO421.4: Design and Modularize the project

CO421.5: Construct, Integrate, Test and Implement the Project.

CO421.6: Prepare the project Documentation and present the Report using appropriate method.

Course Outcomes – Program Outcomes mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1		✓										✓		
C421.2	✓		✓		✓							✓		
C421.3				✓		✓	✓	✓				✓		
C421.4			✓			✓	✓	✓				✓	✓	
C421.5					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
C421.6									✓	✓	✓	✓	✓	

Course Outcomes – Program Outcome correlation

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
C421.1	2	3										2		
C421.2			2		3							2		
C421.3				2		2	3	3				2		
C421.4			2			1	1	2				3	2	
C421.5					3	3	3	2	3	2	2	3	2	1
C421.6									3	2	1	2	3	

Note: The values in the above table represent the level of correlation between CO's and PO's:

1. Low level
2. Medium level
3. High level

Project mapping with various courses of Curriculum with Attained PO's:

Name of the course from which principles are applied in this project	Description of the device	Attained PO
C2204.2, C22L3.2	Gathering the requirements and defining the problem, plan to develop model for detection and classification of OSCC	PO1, PO3, PO8
CC421.1, C2204.3, C22L3.2	Each and every requirement is critically analyzed, the process model is identified	PO2, PO3, PO8
CC421.2, C2204.2, C22L3.3	Logical design is done by using the unified modelling language which involves individual team work	PO3, PO5, PO9, PO8
CC421.3, C2204.3, C22L3.2	Each and every module is tested, integrated, and evaluated in our project	PO1, PO5, PO8
CC421.4, C2204.4, C22L3.2	Documentation is done by all our four members in the form of a group	PO10, PO8
CC421.5, C2204.2, C22L3.3	Each and every phase of the work in group is presented periodically	PO8, PO10, PO11
C2202.2, C2203.3, C1206.3, C3204.3, C4110.2	Implementation is done and the project will be handled by the social media users and in future updates in our project can be done based on detection for Oral Cancer	PO4, PO7, PO8
C32SC4.3	The physical design includes website to check OSCC	PO5, PO6, PO8

ABSTRACT

Oral Squamous Cell Carcinoma (OSCC) is one of the Prevalent form of oral cancer, oftentimes linked to risk factors like tobacco use, alcohol use, chewing betelnut and viral infections. Early diagnosis and treatment of OSCC and other oral mucosal potentially malignant lesions is the clinician's best bet to improve prognosis. In this research paper, we propose the new Deep Learning model of Xception spiking Fractional Neural Network (XSFN-NET) to different OSCC of the histopathology. The process starts with the image intensification with the Medav filter, then accurate pixel-wise segmentation with the Parallel Reverse Attention Network. Feature extraction is done with a combination of hand-crafted methods—Local Vector Pattern (LVP), Discrete wavelet transform (DWT), and Gray-level runlength matrix (GLRM)—and the deep features from the Xception model. The features are the combined with Fractional Calculus (FC) and classified by a Deep Spiking Neural Network (DSNN) to make efficient and biologically inspired decisions. The result of XSFN-Net reached an accuracy of 91%. Its precision was 93.51%, recall was 93.01%, and the F1-score was 93.64%. Overall, XSFN-Net represents a significant step forward in computer-aided diagnosis, offering a powerful and intelligent method for OSCC detection. Its hybrid feature fusion strategy, enhanced preprocessing pipeline, and neuro-inspired classifier contribute to improved diagnostic accuracy and reliability. The methodology and implementation presented in this work open avenues for future advancements in multi-class classification, explainable AI, clinical deployment, and real-time screening applications.

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1.INTRODUCTION

Approximately 84–97% of malignancies of the oral cavity is the oral squamous cell carcinomas, is the common types of the oral cancer. It starts in the squamous cells that line oral cavity and is frequently connected to tobacco use and other lifestyle choices. use, drinking alcohol, chewing betelnuts, and being infected with viruses like HPV.

Early detection of OSCC is crucial because it significantly boosts survival rates. Patients diagnosed early have an 85% five-year survival rate, while those diagnosed at a later stage only have a 40% five-year survival rate. Early detection of OSCC is crucial, as it significantly improves survival rates. If identified in the early stages, patients have a five-year survival rate exceeding 80%, but this drops to below 50% in advanced stages.

Despite advances in cancer research and molecular biology, early detection of OSCC remains extremely challenging. Most patients are diagnosed later in life because early lesions are often subtle and asymptomatic, which lowers survival rates and creates a negative outlook. To detect signs of OSCC, physicians use scans, physical examinations, and reviews of symptoms alongside tissues examinations through biopsies.

Traditional methods face challenges microscopy and staining as well as different levels of experience and artistry of different pathologists, making them tedious and imprecise. This subjectivity is tied to the risks of inaccurate diagnosis and the risks of “anesthesia of the awakened subject” [1].

For OSCC, features are image-based and nonimage-based. They are all blended and processed with a CNN. This approach has the potential to develop low-cost AI algorithms for cancer screening a AI-based algorithms for screening and diagnosis of cancer in underdeveloped and poor regions [2]. This work proposes an ensemble architecture for OSCC classification from histopathological images using the Inception ResNetV2 and Xception models. Using both networks in the ensemble approach improves generalization and feature diversity, two benefits of ensemble approaches that increase accuracy [3].

The model in [5] was designed to simplify deployment for primary care clinics, field healthcare settings, and rural screening camps since it runs on low-power devices. This research introduces a hybrid ensemble model which features a unique Convolutional VGG. The model implements explainable artificial intelligence (XAI) through LIME to boost both prediction accuracy and result interpretability [8]. The model employs gradient-boosted decision trees together with deep feature extraction to merge data-driven learning with structured decision-making. The ensemble models built from EfficientNet architectures demonstrate superior OSCC detection accuracy while maintaining the lowest computational requirements according to [10].

XSFN-Net which functions as a hybrid framework by uniting Deep Spiking Neural Net works (DSNN) with handcrafted features consisting of texture based Local Vector Patterns (LVP), frequency-based Discrete Wavelet Transform (DWT) and statistical descriptors using Gray Level Run Matrix (GLRM). The framework also integrates pixel-level lesion segmentation using PraNet.

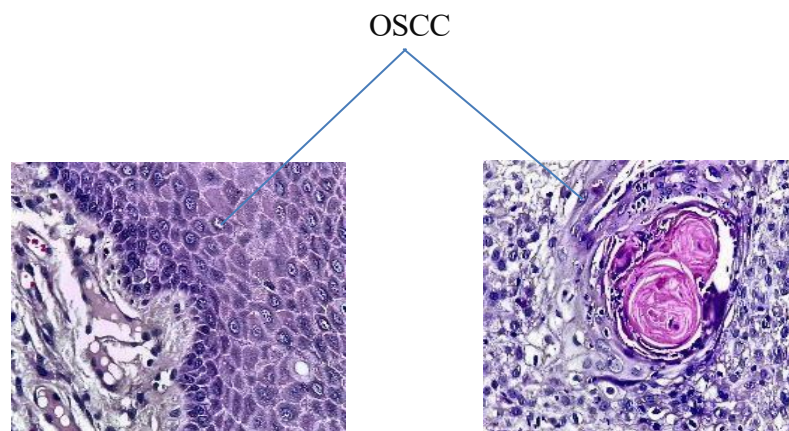


FIG 1. 1 CLASSIFICATION OF NORMAL AND OSCC HISTOPATHOLOGICAL TISSUES

This figure presents the visual comparison between normal oral epithelial tissues and Oral Squamous Cell Carcinoma (OSCC) samples observed under histopathological examination. The normal tissue displays a uniform epithelial cell arrangement, regular nuclear size, and smooth cytoplasmic boundaries, indicating structural harmony. In contrast, OSCC tissues show a significant loss of cellular organization, irregular nuclei, dense chromatin (hyperchromatism), keratin pearl

formation, and stromal invasion, all of which are pathological indicators of malignancy.

The distinction between these tissue types forms the core basis for automated classification in the proposed XSFN-Net model. During model training, these microscopic differences are captured by the Xception-based deep feature extractor, which learns complex spatial and textural variations. Simultaneously, handcrafted feature extraction techniques such as Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray-Level Run-Length Matrix (GLRM) quantify pixel-level variations in texture, frequency, and statistical intensity. The fractional calculus fusion layer integrates both handcrafted and deep features, enhancing discriminative representation between normal and cancerous regions. Finally, the Deep Spiking Neural Network (DSNN) mimics biological neuron activations to classify tissues effectively, achieving high accuracy in distinguishing OSCC from normal samples.

These multi-domain features are fused using Fractional Calculus (FC) to achieve a richer and more continuous representation of tissue characteristics. This fusion process amplifies subtle variations that may be imperceptible to standard convolutional layers. The integrated feature set is then classified by a Deep Spiking Neural Network (DSNN), which mimics the behavior of biological neurons, providing efficient, interpretable, and high-accuracy classification of normal versus OSCC tissues.

Thus, this figure not only establishes the pathological basis for cancer diagnosis but also visually supports how XSFN-Net leverages both structural and textural cues to achieve superior diagnostic precision. It highlights the transition from classical visual interpretation by pathologists to an automated, biologically inspired deep learning framework that can assist in early and reliable OSCC screening.

Overall, this figure highlights the pathological basis of the classification task and visually reinforces how morphological changes in tissue structure are leveraged by the XSFN-Net model for accurate, biologically inspired cancer detection.

1.1 Motivation

Oral Squamous Cell Carcinoma (OSCC) is among the most prevalent and life-threatening malignancies of the oral cavity, accounting for nearly 90–95% of oral cancers worldwide. Despite major advancements in medical imaging and oncology, early-stage diagnosis of OSCC remains a critical challenge, particularly in developing regions. Conventional diagnosis relies heavily on manual histopathological analysis, which is both time-consuming and subjective, often leading to inconsistent results due to variations in staining quality, human fatigue, and the pathologist’s experience.

Microscopic differences between normal and malignant tissues—such as nuclear pleomorphism, hyperchromatism, keratin pearl formation, and tissue disorganization—are subtle and complex. Detecting these differences requires not only clinical expertise but also precision tools capable of quantifying fine-grained patterns across thousands of histopathological images. Traditional computer vision techniques and shallow machine learning models struggle to capture these multi-scale spatial and textural characteristics effectively.

This limitation created the motivation to design an intelligent, automated, and biologically inspired system capable of performing accurate OSCC classification. The proposed XSFN-Net model emerges from this need — combining the power of deep learning and neuro-inspired computation. It integrates deep convolutional feature extraction (Xception model) with handcrafted textural features (LVP, DWT, GLRM) and enhances them through Fractional Calculus-based feature fusion, followed by classification using a Deep Spiking Neural Network (DSNN).

This hybrid approach not only improves classification accuracy but also mimics the decision-making behavior of biological neurons, offering interpretability and energy efficiency. The model aims to assist medical professionals by reducing diagnostic subjectivity, accelerating image screening, and supporting early detection—which significantly increases survival rates among OSCC patients.

1.2 Problem Statement

Oral Squamous Cell Carcinoma (OSCC) remains one of the most aggressive forms of oral cancer, contributing to significant global morbidity and mortality. Despite improvements in diagnostic technologies, early detection of OSCC continues to be a major challenge due to the complexity and variability of histopathological tissue structures. Traditional diagnosis primarily depends on manual microscopic examination of biopsy samples by expert pathologists, a process that is subjective, time-consuming, and prone to inter-observer variability.

Existing automated diagnostic systems that utilize conventional Convolutional Neural Networks (CNNs) or handcrafted feature extraction techniques encounter several inherent limitations in accurately classifying Oral Squamous Cell Carcinoma (OSCC) from histopathological images. These models often exhibit low generalization capability across diverse magnification levels and variations in tissue morphology. Moreover, they provide insufficient feature representation, as traditional deep models tend to overlook the critical micro-textural and frequency-based information that is essential for precise differentiation between normal and malignant tissues. Another major drawback lies in their lack of biological interpretability, which makes it challenging for medical practitioners to understand and trust the automated decision-making process. In addition, the high computational cost and absence of energy-efficient processing mechanisms further restrict their applicability in real-time or resource-constrained clinical environments.

These challenges collectively result in reduced accuracy and reliability in identifying subtle morphological differences in early-stage OSCC lesions. Hence, there is an urgent need for a robust, hybrid, and biologically inspired deep learning architecture capable of integrating multi-domain features — including spatial, textural, and frequency characteristics — to enhance diagnostic performance and interpretability.

1.3 Objective

The primary objective of this research is to design and implement a hybrid, biologically inspired deep learning model capable of accurately classifying Oral Squamous Cell Carcinoma (OSCC) from histopathological images. The proposed XSFN-Net (Xception Spiking Fractional Neural Network) aims to overcome the limitations of traditional diagnostic methods by improving accuracy, interpretability, and computational efficiency in medical image analysis. To achieve this, the study focuses on developing a robust preprocessing framework using the Medav Filter to remove noise, preserve fine tissue structures, and enhance image contrast for better visibility of cellular features. It also aims to perform precise lesion segmentation using the Parallel Reverse Attention Network (PraNet), enabling pixel-level detection of abnormal epithelial regions.

Further, the project seeks to extract multi-domain features by combining handcrafted descriptors—such as Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray-Level Run-Length Matrix (GLRM)—with deep hierarchical features derived from the Xception model to capture both spatial and textural details effectively. These diverse features are then fused using a Fractional Calculus-based feature fusion mechanism, which enhances the richness of feature representation and improves classification accuracy. To achieve biologically inspired and energy-efficient learning, a Deep Spiking Neural Network (DSNN) is employed to simulate neuron-like processing behavior for improved interpretability and robustness.

The project further aims to evaluate the proposed XSFN-Net model against existing architectures such as CNN and OralNet using standard performance metrics like accuracy, precision, recall, and F1-score to validate its superiority. Additionally, a user-friendly interface will be developed to visualize and interpret the model's classification outcomes, facilitating clinical applicability and assisting pathologists in early and reliable diagnosis of OSCC. Through these objectives, the research intends to contribute a powerful, explainable, and efficient AI-driven diagnostic framework for the early detection of oral cancer.

2. LITERATURE SURVEY

Oral Squamous Cell Carcinoma (OSCC) represents the most prevalent form of oral malignancy, accounting for nearly 90–95% of oral cancer cases worldwide. Despite advancements in histopathology and imaging, early detection remains a major challenge due to the morphological complexity and subtlety of early lesions. Manual diagnosis using biopsy slides is the gold standard, but it is highly subjective and depends on the pathologist’s expertise, leading to potential inconsistencies and diagnostic delays. Histopathological images contain valuable visual information, including cellular arrangement, nuclear pleomorphism, chromatin distribution, and tissue organization, which can serve as crucial indicators of malignancy. However, manual interpretation of these features is laborious and prone to inter-observer variability. Consequently, computer-aided diagnostic (CAD) systems have emerged to assist pathologists by automating feature extraction, classification, and decision-making.

Automated analysis of histopathological images for cancer detection has rapidly advanced with deep learning, yet domain-specific challenges remain. Systematic reviews of deep learning in oral cancer show that CNN-based approaches achieve promising accuracy but face variability across datasets, staining protocols, and magnification levels; these reviews recommend combining robust preprocessing, segmentation, and multi-scale feature extraction to improve clinical reliability.

Accurate lesion segmentation is a critical precursor to reliable classification. Parallel Reverse Attention Network (PraNet) introduced a practical and effective approach to segmentation by aggregating high-level features via a parallel partial decoder and refining boundaries with a reverse-attention module; PraNet demonstrated strong generalizability and real-time performance on polyp segmentation tasks, and has since been adapted for medical image segmentation problems that require precise boundary localization.

Integrating such segmentation models in OSCC detection enables pixel-wise lesion localization, which helps isolate cancerous regions before feature extraction. By incorporating PraNet in preprocessing, the XSFN-Net model achieves accurate

and focused segmentation of histopathological structures, ensuring that only relevant pathological regions are analyzed.

Ramya et al. [1] created XSFN-Net which functions as a hybrid framework by uniting Deep Spiking Neural Networks (DSNN) with handcrafted features consisting of texture based Local Vector Patterns (LVP), frequency-based Discrete WaveletTransform (DWT) and statistical descriptors using GrayLevelRunMatrix (GLRM). The framework also integrates pixel-level lesion segmentation using PraNet.

G.A.I. Devindi et al. [2] developed a Multimodal deep convolutional neural network pipeline aimed at the early detection of the oral cancer. Their model utilizes images of the oral cavity captured via smartphone, alongside patient information such as age, gender, and lifestyle habit to enhance prediction accuracy. While the early-fusion multimodal process achieved only 81%, it reflects genuine clinical practice workflows, leading to a more complete and accessible diagnostic model which is more practical in a healthcare setting.

The introduction of Convolutional Neural Networks (CNNs) revolutionized medical image classification by automatically learning hierarchical features from raw images. Deep architectures such as VGGNet, ResNet, Inception, DenseNet, and Xception became widely adopted due to their ability to extract discriminative spatial features with minimal manual intervention.

Madhusmita Das et al. [3] developed an ensemble deep learning model for OSCC detection, employing a two-stage technique to select and combine the best-performing pretrained models—InceptionResNetV2 and Xception. This architecture capitalizes on ensemble learning’s strength in improving prediction performance, even without relying on handcrafted features or custom segmentation. Researchers from Amity University [4] focused on developing a lightweight OSCC detection system using CNN and TensorFlow Lite. Although the paper lacks comprehensive performance metrics, it demonstrates progress toward creating mobile and edge compatible solutions for low-cost, real-time screening in remote or resource-constrained areas.

S. Vasuki et al. [5] proposed an ensemble deep learning framework that combines modified VGG networks and CNNs for feature extraction, followed by classification using XGBoost. This hybrid approach effectively captures nonlinear feature relationships, improving both classification accuracy and model generalization. Desai et al. [6] introduced a lightweight diagnostic model using DenseNet201 and FixCaps applied to mobile-captured oral cavity images. Their work focuses on early screening of OSCC and potentially malignant disorders, offering a portable and effective solution for community-level diagnostics.

These ensemble approaches demonstrated the effectiveness of feature fusion in improving classification performance. However, most models still operate using integer-order feature fusion methods, which can limit their ability to integrate fine-grained details from heterogeneous feature sources. Moreover, such models are computationally expensive and lack biological interpretability, making them unsuitable for real-time diagnostic use.

Somyanonthanakul et al. [7] extended the research scope by introducing a Fuzzy Deep Learning (FDL) model that incorporates both clinical and pathological data to predict.

Samiha Islam et al. [8] underscored the need for explain ability for clinical AI applications with the combination of EfficientNetB3, along with LIME (Local Interpretable Model agnostic Explanations) which provided visual rationalization for model predictions. B. Ananthakrishnan et al. [9] proposed a Deep Neural Network-based pipeline for the automatic detection and classification of the OSCC with histopathological images; also achieved high sensitivity and specificity on test sets.

Albalawi et al. [10] validated the capabilities of EfficientNet architectures for OSCC detection using standard histopathological data sets. Their study revealed both high classification and classification robustness, confirming the model's efficiency.

In the proposed XSFN-Net framework, FC is utilized to fuse deep and handcrafted features into a unified representation space. This fractional fusion ensures a richer, smoother, and more continuous feature representation, preserving both local and global image characteristics. Studies in other medical domains have demonstrated that FC-based fusion improves robustness against noise, illumination changes, and staining inconsistencies — all of which are prevalent challenges in histopathological analysis.

In recent years, Deep Spiking Neural Networks (DSNNs) have emerged as biologically inspired alternatives to conventional deep neural networks. Unlike traditional neurons that compute continuous activations, spiking neurons communicate through discrete spikes, emulating the way the human brain processes visual information. DSNNs are capable of temporal information encoding, low-power computation, and event-driven processing, making them ideal for medical imaging applications that demand real-time analysis with limited computational resources.

In the context of OSCC detection, DSNNs offer an energy-efficient yet accurate solution by combining deep learning’s representational power with the interpretability of brain-inspired computation. The XSFN-Net leverages DSNNs in its classification phase, allowing biologically plausible decision-making while significantly reducing computational cost compared to traditional CNN classifiers.

From the review of existing literature, several key gaps have been identified. First, most conventional CNN-based methods fail to capture both spatial and textural features effectively, resulting in reduced classification accuracy. Second, traditional handcrafted methods lack robustness and generalization across varying magnification levels and datasets. Third, few existing models incorporate biologically inspired architectures that mimic neural spiking behavior, limiting interpretability and energy efficiency. Fourth, feature fusion techniques in current studies often rely on simple concatenation or integer-order methods, which may not fully exploit the complementary nature of deep and handcrafted features.

These challenges collectively justify the development of a hybrid architecture like XSFN-Net, which integrates Medav filter-based preprocessing, PraNet segmentation, multi-domain handcrafted and deep features, Fractional Calculus- based fusion, and DSNN classification to achieve superior diagnostic performance. This fusion of biologically inspired learning, deep feature extraction, and advanced mathematical modeling bridges the gap between computational efficiency and clinical interpretability.

The reviewed studies demonstrate a clear evolution from manual diagnostic techniques to deep and hybrid learning frameworks in medical image analysis. While earlier methods relied heavily on handcrafted texture descriptors, modern deep learning models have significantly advanced feature learning capabilities. However, the challenges of interpretability, computational cost, and robustness

The proposed XSFN-Net model, by integrating Xception-based deep feature extraction, handcrafted textural features, fractional calculus-based fusion, and spiking neural computation, aims to overcome these limitations. It represents a new direction in OSCC detection research — combining the precision of machine intelligence with the interpretability and efficiency inspired by biological neural systems.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the existing diagnostic systems for Oral Squamous Cell Carcinoma (OSCC), classification of histopathological images primarily depends on manual examination by pathologists and traditional machine learning-based methods. Manual diagnosis involves microscopic inspection of biopsy slides to identify structural abnormalities such as nuclear pleomorphism, keratin pearls, mitotic figures, and loss of epithelial organization. Although effective, this process is highly subjective, time-consuming, and requires considerable expertise.

To assist pathologists, early computer-aided systems used handcrafted feature extraction methods such as Gray-Level Co-occurrence Matrix (GLCM), Local Binary Pattern (LBP), Gray-Level Run-Length Matrix (GLRM), and Discrete Wavelet Transform (DWT). These methods extracted statistical and texture features, which were then classified using Support Vector Machines (SVM), Random Forests (RF), or Artificial Neural Networks (ANNs). While these approaches provided partial automation, they relied heavily on manually selected features and predefined rules, limiting their adaptability to diverse datasets.

Later, Convolutional Neural Network (CNN) architectures were introduced for OSCC image classification. Models like VGG, ResNet, and DenseNet demonstrated improved performance through deep hierarchical learning. However, CNN-based systems suffered from overfitting on small medical datasets, loss of fine-grained texture details, and lack of interpretability. Moreover, most existing deep models required high computational resources, making them unsuitable for deployment in real-time.

The below figure illustrates the workflow of a traditional Oral Squamous Cell Carcinoma (OSCC) detection system based on conventional image processing and machine learning techniques. The process consists of four main stages: image acquisition, preprocessing, feature extraction, and classification.

FLOW CHART OF EXISTING OSCC DETECTION SYSTEM

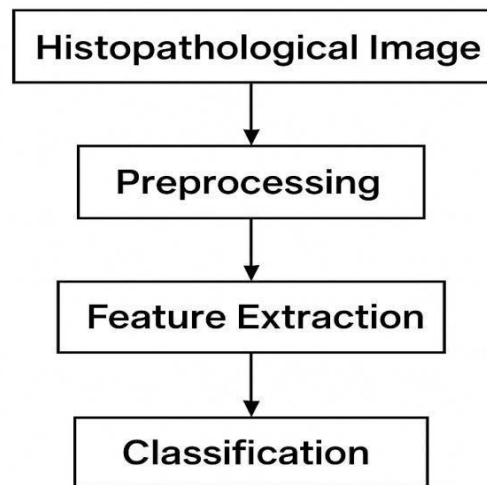


FIG 3.1. FLOW CHART OF EXISTING OSCC DETECTION SYSTEM CLASSIFICATION

The system begins with the collection of histopathological images of oral tissue samples. These images are obtained using a microscope at different magnifications (commonly 100× or 400×). The dataset includes both normal epithelial tissue and OSCC-affected tissue images, which serve as input for the classification model. In preprocessing stage, the acquired images undergo preprocessing to enhance visual quality and remove noise or irrelevant background components. Common techniques include contrast enhancement, image resizing, and filtering (e.g., Gaussian or median filters). The goal is to normalize the images for consistent texture and intensity across the dataset, thereby improving the accuracy of subsequent feature extraction.

After preprocessing, relevant features are extracted to characterize the texture, shape, and color variations of tissues. Traditional systems rely on handcrafted feature extraction methods such as Gray-Level Co-occurrence Matrix (GLCM), Local Binary Pattern (LBP), and Discrete Wavelet Transform (DWT). These techniques compute statistical and spatial relationships between pixels, capturing textural details such as contrast, entropy, and homogeneity. However, they are limited in capturing complex spatial hierarchies in histopathological images.

3.1.1 DISADVANTAGES OF THE EXISTING SYSTEM FOR OSCC CLASSIFICATION

Despite several advancements in image processing and machine learning, the existing systems for Oral Squamous Cell Carcinoma (OSCC) classification still face multiple challenges:

- **Dependence on Large Labeled Datasets:**

Deep Learning models, particularly CNNs, require extensive labeled datasets for effective training. However, obtaining large-scale annotated medical images is challenging due to data privacy concerns and the need for expert annotations.

- **Low Generalization Ability:**

Existing models trained on limited datasets struggle to generalize across images with different magnifications, staining variations, and imaging conditions. As a result, their performance drops significantly when tested on new or unseen data from different laboratories or imaging devices.

- **Dependence on Manual Processing:**

Most existing systems require manual segmentation and preprocessing, which makes the workflow time-consuming and prone to human error. Manual intervention also introduces subjectivity and reduces the repeatability of the diagnostic process.

- **High Computational Complexity:**

Deep learning architectures like ResNet or VGG require large computational resources and long training times. Their heavy reliance on high-end GPUs makes real-time deployment difficult, particularly in clinical or rural healthcare environments where computational resources are limited.

- **Lack of Biological Interpretability:**

Conventional CNN-based systems operate as black-box models, providing high accuracy but offering little to no explanation for their classification decisions. This lack of interpretability reduces clinician trust and limits integration into diagnostic workflows.

- **Inability to Handle Noisy or Low-Contrast Images:**

Many histopathological images contain noise, uneven lighting, or blurred cellular boundaries due to staining inconsistencies. Existing methods often fail to compensate for these issues, leading to inaccurate feature extraction and false classifications.

3.2 PROPOSED SYSTEM

The below flowchart illustrates the workflow of the proposed XSFN-Net (Xception Spiking Fractional Neural Network) for OSCC classification.

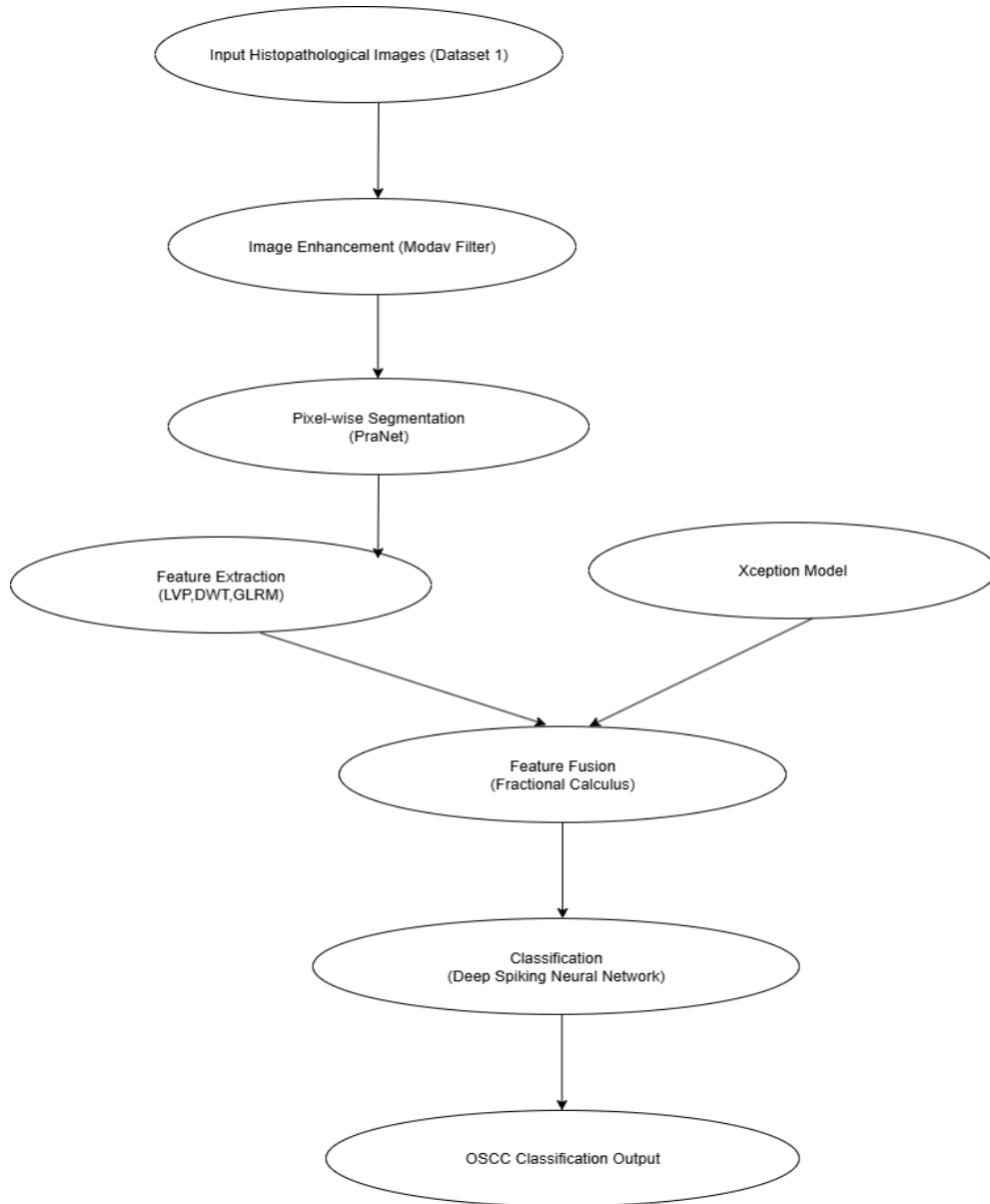


FIG 3.2. FLOW CHART OF PROPOSED SYSTEM

The process begins with histopathological image acquisition, followed by Medav Filter preprocessing, which removes noise and enhances image clarity. The enhanced image is then processed using the PraNet segmentation model to accurately isolate tumor regions. From these segmented regions, both handcrafted features (LVP, DWT, GLRM) and deep features (Xception) are extracted to capture

texture, frequency, and spatial information. These multi-domain features are then combined using Fractional Calculus-based Feature Fusion, creating a rich and discriminative feature representation. Finally, the fused features are classified using a Deep Spiking Neural Network (DSNN), which mimics biological neuron behavior for efficient and interpretable classification. The system outputs whether the input tissue is Normal or OSCC, providing an accurate and energy-efficient solution for oral cancer detection. the proposed XSFN-Net achieves robust performance with high accuracy, interpretability, and computational efficiency.

Advantages over exsisting system:

1. **Enhanced Accuracy:** The proposed XSFN-Net achieves higher classification accuracy by combining handcrafted (LVP, DWT, GLRM) and deep (Xception) features for a more comprehensive tissue representation.
2. **Improved Image Quality:** The Medav Filter effectively removes noise and enhances contrast, preserving important histopathological structures such as nuclei and epithelial boundaries.
3. **Accurate Lesion Segmentation:** The PraNet segmentation model accurately isolates OSCC-affected regions using parallel partial decoding and reverse attention mechanisms, improving diagnostic precision.
4. **Energy-Efficient Classification:** The Deep Spiking Neural Network (DSNN) offers biologically inspired, event-driven computation that significantly reduces energy consumption compared to conventional deep learning models.
5. **Better Generalization:** The hybrid design ensures robust performance across different magnification levels and staining conditions, improving model adaptability to diverse datasets.
6. **High Interpretability:** The biologically inspired architecture of DSNN improves transparency, allowing clinicians to better understand the model's diagnostic reasoning.
7. **Reduced Computational Cost:** Optimized feature fusion and spiking computation reduce memory usage and training complexity.

3.3 FEASIBILITY STUDY

A feasibility study evaluates the practicality and effectiveness of implementing the proposed XSFN-Net system for Oral Squamous Cell Carcinoma (OSCC) detection. Below is a detailed feasibility analysis considering technical, operational, and economic aspects.

1. Technical Feasibility

- **Efficient Image Preprocessing and Segmentation:**

Image preprocessing using the Medav Filter and segmentation using PraNet are highly compatible with modern GPU-based systems, ensuring efficient and accurate data preparation for classification.

- **Advanced Feature Fusion Capability:**

The integration of Fractional Calculus-based Feature Fusion enhances the system's ability to combine deep and handcrafted features, providing a more discriminative feature representation for improved accuracy.

- **Feasibility of Deep Spiking Neural Networks (DSNN):**

The DSNN architecture can be easily implemented using existing computational frameworks, enabling biologically inspired, event-driven classification with low power consumption.

- **Moderate Data Requirements:**

The system can be trained effectively on moderate-sized histopathological datasets (hundreds to thousands of images), eliminating the need for very large datasets or specialized data processing infrastructure.

- **Hardware Compatibility:**

The proposed model performs efficiently on standard GPU hardware confirming that no high-end or cost-prohibitive equipment is required.

2. Operational Feasibility

- **User-Friendly Operation:**

The proposed XSFN-Net system is designed to be fully automated and user-friendly, requiring minimal technical expertise for operation, making it easily usable by pathologists and medical staff.

- **Seamless Integration:**

The model can be smoothly integrated into existing digital pathology workflows and laboratory information systems, enhancing diagnostic efficiency without major infrastructure changes.

- **Ease of Deployment:**

Since the system operates efficiently on standard computational setups, it can be easily deployed in hospitals, research laboratories, and diagnostic centers without specialized technical support.

- **Reduced Processing Time:**

The system performs rapid image processing and classification, supporting real-time or near-real-time diagnostic decisions, which is crucial for clinical workflows.

3. Economic Feasibility

- **Low Development Cost:**

The proposed XSFN-Net system is designed using open-source tools and frameworks such as Python, TensorFlow, and Keras, which eliminates the need for expensive software licenses and reduces overall development costs.

- **Energy-Efficient Architecture:**

The inclusion of a Deep Spiking Neural Network (DSNN) enables low-power, event-driven computation, significantly reducing the system's operational energy consumption compared to traditional deep CNN models.

- **Reduced Diagnostic Costs:**

By automating the classification of OSCC through AI-driven analysis, the system minimizes the need for manual interpretation by pathologists, thereby saving time and reducing labor expenses.

- **Long-Term Investment:**

Although initial development and integration costs may be high, the long-term benefits of improved diagnostic accuracy and efficiency justify the investment. The system's ability to adapt to new data and conditions ensures sustainable benefits.

3.4 USING COCOMO MODEL

The COCOMO (Constructive Cost Model) is a widely used estimation technique in software engineering that helps predict the effort, development time. Based on the

project's research-oriented and modular nature, the proposed XSFN-Net system falls under the "Organic Mode" of the COCOMO model, as it involves small to medium-sized development teams, clear objectives, and well-understood requirements.

The Basic COCOMO model relies on three key formulas for estimating effort, development time, and the number of people required:

- Effort (E) = $a \times (\text{KLOC})^b$ (measured in Person-Months)
- Development Time (T) = $c \times (E)^d$ (measured in Months)
- People Required (P) = E / T

In these formulas, KLOC refers to the estimated number of lines of code in thousands, and the constants (a, b, c, d) vary based on the project type. For Semi-Detached projects, the constants are $a = 2.4$, $b = 1.05$, $c = 2.5$, and $d = 0.38$.

The estimated code size for the XSFN-Net system (including preprocessing, segmentation, feature extraction, fusion, and classification modules) is approximately 4.5 KLOC. Using the formula:

$$E = 2.4 \times (4.5)^{1.05} = 11.4 \text{ person-months (approximately)}$$

the total development duration is estimated to be around 6 months, assuming a small research team

$$D = 2.5 \times (11.4)^{0.38} = 6.3 \text{ months (approximately)}$$

The project would ideally require a team of 4 members, which may include machine learning engineers, backend developers, frontend developers, and testers to ensure timely and efficient development. Several factors can influence these estimates, such as the complexity of the machine learning model, the size and quality of the OSCC dataset, the intricacies of integrating the Flask backend with the frontend interface, and the time allocated for thorough testing and validation. While the COCOMO model provides a solid framework for initial estimation, real-world development may require adjustments based on faced during the project lifecycle. Based on the COCOMO analysis, the proposed XSFN-Net system is time-feasible, with development achievable within 6 months using a small, well-coordinated research team. The modular design and use of pre-trained architectures further reduce development time while maintaining high system performance.

4. SYSTEM REQUIREMENTS

4.1 SOFTWARE REQUIREMENTS

- | | |
|-------------------------|---|
| 1. Operating System | : Windows 11, 64-bit Operating System |
| 2. Hardware Accelerator | : 12th Gen Intel(R) Core (TM) i5-1235U (1.30 GHz) |
| 3. Coding Language | : Python |
| 4. Python distribution | : Google Colab Pro, Flask |
| 5. Browser | : Any Latest Browser like Chrome |

4.2 REQUIREMENT ANALYSIS

The Requirement Analysis phase defines the hardware, software, and functional needs necessary for the successful design and implementation of the proposed XSFN-Net (Xception Spiking Fractional Neural Network) for Oral Squamous Cell Carcinoma (OSCC) classification. This stage ensures that the system's objectives are achievable, the tools are compatible, and the workflow is optimized for efficiency, scalability, and accuracy. The backend is developed using Python with Flask, while the frontend utilizes HTML, CSS, and JavaScript for user interaction.

Non-functional requirements ensure that the system is reliable, efficient, and user-friendly. The XSFN-Net model is designed to deliver high classification accuracy ($\geq 90\%$) with real-time inference capability, ensuring its clinical relevance. The architecture is scalable, allowing it to adapt to larger datasets or other histopathological image types. It is also interpretable, as the DSNN offers biologically meaningful insights into the classification process. The modular design enhances maintainability, enabling future updates without major reconfiguration, while secure data handling ensures patient data confidentiality and compliance with ethical standards.

The proposed XSFN-Net system requires moderate computational resources and leverages open-source software tools to deliver a cost-effective and high-performance solution. Its well-defined functional and non-functional requirements ensure successful development and deployment for automated, interpretable, and efficient OSCC detection in both research and clinical settings.

4.3 HARDWARE REQUIREMENTS:

1. System Type : 64-bit operating system, x64-based processor
2. Cache memory : 4MB(Megabyte)
3. RAM : 16GB (gigabyte)
4. Hard Disk : 500GB SDD
5. GPU : Intel® Iris® Xe Graphics

4.4 SOFTWARE

The OSCC detection project leverages a comprehensive set of software tools, technologies, and system configurations to ensure accurate, efficient, and scalable development and deployment. The system is designed to operate on Windows 11, 64-bit Operating System, ensuring compatibility with modern hardware and software environments. The project utilizes the CPU as the primary hardware accelerator, providing sufficient computational power for model training, inference, and backend processing.

The core development is carried out using the Python programming language, known for its simplicity, flexibility, and extensive library support. The Python environment is managed using Google Colab Pro for initial model development and training, which offers access to enhanced computational resources and faster processing speeds. For backend development and web application deployment, the Flask framework is employed, enabling efficient handling of API requests, data processing, and seamless integration with the frontend.

For the frontend, HTML5 and CSS3 are used to design and style the web interface, while Bootstrap ensures responsiveness and compatibility across devices. This guarantees an intuitive and user-friendly experience for end-users. Custom CSS styles are also applied to enhance the visual appeal and maintain a consistent layout. The application is compatible with any modern web browser, including the latest versions of Google Chrome, Mozilla Firefox, and Microsoft Edge, ensuring cross-browser accessibility and performance.

In terms of machine learning, the project employs TensorFlow/Keras to develop a Convolutional Neural Network (CNN) that efficiently extracts features from MRI images for tumor detection. Additionally, scikit-learn is used to implement a Support

Vector Machine (SVM) classifier, enhancing the accuracy and robustness of the classification process.

For image preprocessing, OpenCV is utilized to handle tasks such as image resizing, format validation, and noise reduction, ensuring that input data is clean and standardized for better prediction accuracy. NumPy supports efficient numerical computations, especially for handling large datasets and image matrices. Development and model evaluation are conducted within Jupyter Notebook, providing an interactive environment for experimentation and refinement. Visualization tasks, such as plotting accuracy curves and confusion matrices, are handled using Matplotlib, aiding in model performance analysis.

Overall, the integration of these software tools, along with the specified system requirements, ensures that the OSCC detection system is accurate, efficient, scalable, and compatible with modern computing environments.

4.5 SOFTWARE DESCRIPTION

The OSCC detection system requires a modern and stable operating system, with Windows 11, 64-bit being the recommended choice. This ensures compatibility with the latest development tools, security updates, and efficient system performance. The CPU will serve as the primary hardware accelerator, which is suitable for running pre-trained models and managing backend operations. For large-scale training or intensive computations, cloud platforms like Google Colab Pro are utilized, offering access to advanced GPUs and faster processing capabilities.

The project is developed using Python, a versatile and widely-used programming language known for its extensive libraries and frameworks that simplify the development of machine learning and deep learning applications. The Python distribution includes platforms like Google Colab Pro for efficient model training and Flask for creating a lightweight and robust backend web application. Flask is particularly effective for deploying machine learning models as web services, allowing seamless integration between the backend and frontend.

Additionally, a web browser such as Google Chrome or any other latest version browser is required for accessing and interacting with the Flask-based web application.

5. SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The system architecture of the proposed XSFN-Net (Xception Spiking Fractional Neural Network) provides a structured and modular framework for the automatic classification of Oral Squamous Cell Carcinoma (OSCC) from histopathological images. It is designed to integrate multiple computational stages — image preprocessing, segmentation, feature extraction, feature fusion, and classification — into a unified hybrid deep learning pipeline. Each stage performs a specific function that contributes to accurate, interpretable, and energy-efficient OSCC detection.

The acquired images are then processed in the preprocessing module, where the Medav Filter (a combination of adaptive median and mean filters) is applied to remove noise, normalize brightness, and enhance contrast while preserving fine cellular structures. This ensures that images are consistent in quality and suitable for downstream processing. The next stage is the segmentation module, implemented using the Parallel Reverse Attention Network (PraNet). This component performs pixel-level segmentation to isolate OSCC-affected regions from surrounding healthy tissues. PraNet utilizes a Parallel Partial Decoder (PPD) for high-level feature aggregation and a Reverse Attention (RA) mechanism for fine boundary refinement. This segmentation step ensures that only relevant pathological regions are analyzed, reducing background interference and improving the precision of feature extraction.

Following segmentation, the system proceeds to the feature extraction module, which operates through two parallel streams — handcrafted and deep feature extraction. The handcrafted stream extracts textural and statistical features using Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray-Level Run-Length Matrix (GLRM) techniques. These methods capture fine-grained texture, frequency, and intensity variations that reflect tissue morphology. In parallel, the deep learning stream employs the Xception network, which uses depthwise separable convolutions to learn hierarchical spatial features, enabling the identification of complex cellular structures and tissue patterns.

The outputs from both feature extraction streams are then integrated in the feature fusion module, where Fractional Calculus-based Feature Fusion (FCF) is applied. Fractional calculus enables smooth interpolation between differentiation and integration, allowing the system to effectively combine handcrafted and deep features while retaining both local texture and global spatial information.

Finally, the output module displays the classification result and performance metrics such as accuracy, sensitivity, specificity, and confusion matrix visualization. This allows users, particularly pathologists and clinicians, to interpret the outcomes easily and make informed diagnostic decisions.

Overall, the XSFN-Net architecture follows a hybrid, modular, and biologically inspired design. It combines the precision of deep learning, the interpretability of handcrafted features, and the energy efficiency of spiking neural computation. This architecture ensures robust performance, scalability, and clinical applicability, making it a promising solution for automated and interpretable OSCC diagnosis in histopathological imaging.

5.1.1 DataSet

The proposed XSFN-Net system uses a histopathological image dataset for the classification of Oral Squamous Cell Carcinoma (OSCC). The dataset contains high-resolution microscopic images of normal and cancerous oral epithelial tissues, stained using Hematoxylin and Eosin (H&E) to clearly highlight nuclei and cytoplasmic regions. The images are collected from public medical repositories such as Kaggle and The Cancer Imaging Archive (TCIA), ensuring authenticity and expert annotation by pathologists.

Each image is captured at two magnification levels — $100\times$ and $400\times$ — to provide both global and detailed tissue features. The dataset typically includes around 1224 images, evenly distributed between normal and OSCC-affected classes, ensuring class balance. For model training and evaluation, the dataset is divided into 70% training, 20% validation, and 10% testing sets.

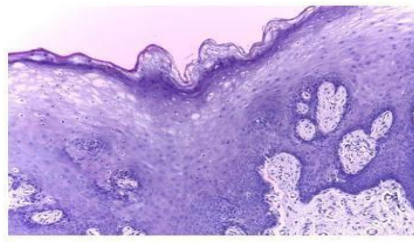
Before feeding into the model, all images are resized to 224×224 pixels and normalized for intensity consistency. The Medav Filter is applied during preprocessing to remove noise and enhance contrast. This dataset provides sufficient diversity in color, staining, and texture variations, allowing the model to learn robust features and achieve high generalization performance.

Feature	Details
Total Images	1224
Total Patients	95
Tissue Classes	Normal and OSCC
Class Distribution	Normal: 612, OSCC: 612
Image Type	Hematoxylin and Eosin (H&E) stained histopathological images
Applications	Classification and detection of Oral Squamous Cell Carcinoma using the proposed hybrid deep learning model (XSFN-Net)

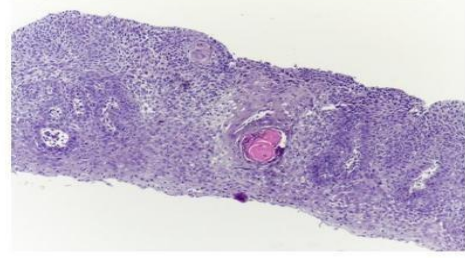
TABLE 1. DATASET DESCRIPTION

Tissue Classes:

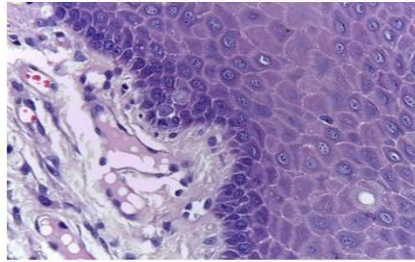
- **Normal Tissue (612 images):** These images represent healthy oral epithelial tissues with regular cellular arrangements, uniform nuclei, and intact epithelial layers. The cells appear well-differentiated with no signs of dysplasia or keratinization. These images serve as the baseline reference for comparison against cancerous samples.
- **OSCC Tissue (612 images):** These images represent oral epithelial tissues affected by squamous cell carcinoma. They exhibit distinct malignant features such as nuclear pleomorphism, hyperchromatism, irregular epithelial architecture, keratin pearls, and abnormal mitotic figures. These features indicate uncontrolled cellular proliferation and invasion into the underlying connective tissue, typical of OSCC pathology.



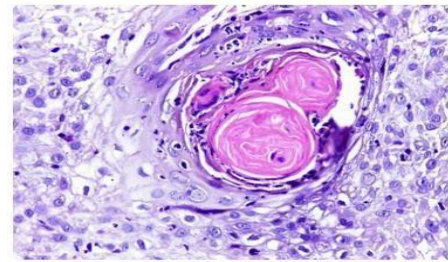
Normal 100x magnification



OSCC 100x magnification



Normal 400x magnification



OSCC 400x magnification

FIG 5.1 DIFFERENT OSCC DATA SET IMAGES.

Image Characteristics:

- Images are captured at 100× and 400× magnifications to capture macroscopic tissue organization and microscopic cellular details.
- All images are resized to 224×224 pixels to ensure uniform input dimensions for the XSFN-Net model.
- H&E staining provides a distinct color contrast — nuclei appear in shades blue to purple (Hematoxylin), and the cytoplasm and connective tissue appear in pink to red (Eosin).

Applications:

- The proposed XSFN-Net system can automatically classify normal and OSCC histopathological images, assisting pathologists in accurate and efficient cancer diagnosis.
- Enables early identification of malignant changes in oral tissues, improving prognosis and patient survival rates through timely treatment.

5.1.2 DATA PRE-PROCESSING

Data preprocessing is a crucial step in preparing histopathological images for efficient and accurate classification. It ensures image quality, uniformity, and suitability for feature extraction and deep learning model training. The preprocessing pipeline in the proposed **XSFN-Net system** includes several stages as described below:

Noise Removal using Medav Filter: The Medav Filter, a hybrid combination of adaptive median and mean filters, is applied to remove noise and smoothen image texture

Contrast Enhancement: Histogram equalization and intensity normalization are performed to enhance the visibility of histological structures and ensure consistent lighting and contrast across all samples.

Segmentation using PraNet: The Parallel Reverse Attention Network (PraNet) is employed for pixel-wise segmentation of lesion regions.

- **Parallel Partial Decoder (PPD)** extracts high-level semantic features.
- **Reverse Attention (RA)** modules refine boundaries and focus on tumor regions.

This isolates the pathological area, ensuring that only relevant tissue regions are analyzed during feature extraction.

Normalization: Image pixel values are normalized to a range of $[0, 1]$ or standardized with zero mean and unit variance to stabilize training and improve model convergence.

Data Augmentation: Techniques such as rotation, flipping, scaling, brightness adjustment, and translation are applied to artificially increase the dataset size, reduce overfitting, and improve model generalization.

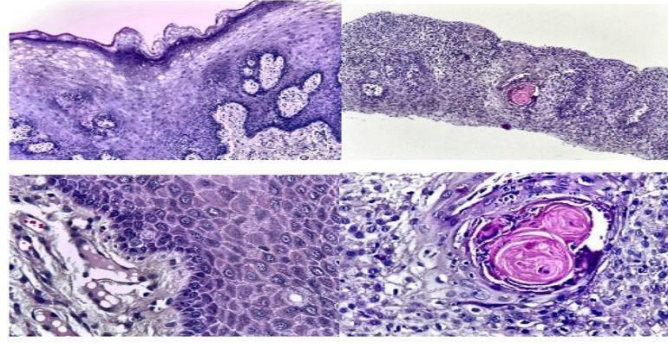


FIG 5.2 IMAGE AFTER APPLYING THE PREPROCESSING TECHNIQUE.

5.1.3 FEATURE EXTRACTION

Feature extraction is a critical step in the proposed XSFN-Net system, as it captures distinctive characteristics of histopathological images that differentiate normal and OSCC-affected tissues.

Handcrafted feature extraction plays a vital role in the proposed XSFN-Net system, as it captures the fine-grained texture, statistical, and structural characteristics of histopathological images that are essential for distinguishing normal and OSCC (Oral Squamous Cell Carcinoma) tissues. Unlike purely deep learning-based methods, handcrafted techniques analyze the micro-level spatial relationships and pixel intensity distributions, providing interpretable and biologically relevant features. In this project, three robust texture-based methods are utilized: Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray Level Run Length Matrix (GLRM).

The Local Vector Pattern (LVP) method extracts directional and intensity-based local features from neighborhood pixels, effectively capturing micro-textural variations present in epithelial tissues. The Discrete Wavelet Transform (DWT) is applied to decompose the image into multiple frequency sub-bands (LL, LH, HL, HH), allowing analysis of both spatial and frequency domains simultaneously. The Gray Level Run Length Matrix (GLRM) technique computes statistical texture features by analyzing the number of consecutive pixels (runs) having the same gray-level intensity in specific directions.

Key Features Extracted from Handcrafted Feature Extraction:

These statistical features are extracted primarily from GLRM (Gray Level Run Length Matrix) and DWT (Discrete Wavelet Transform) outputs — they quantify texture, structure, and intensity variations in histopathological images.

- **Low Gray Level Run Emphasis (LGRE):** Highlights the occurrence of short runs with low gray-level values, emphasizing dark regions (nuclei-rich areas).
- **High Gray Level Run Emphasis (HGRE):** Highlights bright pixel regions, helping differentiate cytoplasmic zones.
- **Energy:** It represents the combination of the class features in the DWT, which means that the text is identical.

$$\text{energy} = \sum_{i,j} R(i,j)^2$$

- **Contrast:** Difference between the highest and lowest intensity levels; identifies edges and boundaries.

$$C = \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} (i - j)^2 P(i, j)$$

5.1.4 MODEL BUILDING:

The proposed XSFN-Net (Xception Spiking Fractional Neural Network) is a hybrid deep learning architecture specifically designed to classify Oral Squamous Cell Carcinoma (OSCC) and normal histopathological tissue images with high accuracy, interpretability, and computational efficiency. The model building process involves several key stages, including multi-domain feature extraction, fractional calculus-based feature fusion, and biologically inspired spiking neural network-based classification. By combining handcrafted and deep features, XSFN-Net leverages both traditional statistical descriptors and hierarchical deep representations to overcome the limitations of existing CNN-based models.

In parallel, the deep feature extraction branch utilizes the Xception convolutional neural network, which is built upon the principle of depthwise separable convolutions. This design reduces computational cost while maintaining high feature learning capability. The Xception network extracts high-level spatial and contextual features, such as nuclear pleomorphism, epithelial disorganization, and keratinization patterns characteristic of OSCC. The network is pretrained on the ImageNet dataset and fine-tuned using OSCC histopathological images to ensure

domain-specific feature adaptation. The outputs from the final convolutional layers are flattened into a feature vector representing deep semantic tissue information.

The outputs from both feature branches are then passed to the Fractional Calculus-Based Feature Fusion (FCFF) module, which performs an intelligent combination of handcrafted and deep feature vectors. Unlike simple concatenation, fractional calculus introduces a fractional-order integration mechanism that adjusts the contribution of each feature type, achieving a smooth balance between local texture details and global contextual patterns. This fusion process produces a hybrid discriminative feature vector that encapsulates multiple image representations — spatial, statistical, and frequency-based — enabling the model to better distinguish subtle variations between healthy and malignant tissue structures.

The fused feature vector is then fed into the classification module, which is implemented using a Deep Spiking Neural Network (DSNN). The DSNN mimics biological neuron behavior by transmitting information through discrete spikes rather than continuous activations. Each neuron follows the Leaky Integrate-and-Fire (LIF) model, which accumulates incoming spikes and fires when a threshold potential is reached. This event-driven computation makes the DSNN energy-efficient and capable of learning spatiotemporal patterns that are biologically meaningful. The DSNN classifier analyzes the fused feature representations and outputs a prediction indicating whether the input image belongs to a Normal or OSCC class. The use of spiking neurons not only reduces power consumption but also enhances the interpretability of the classification process by correlating spikes with discriminative visual features.

XSFN-NET MODEL:

The proposed XSFN-Net (Xception Spiking Fractional Neural Network) architecture is a hybrid deep learning model that integrates Xception-based convolutional layers, Fractional Calculus-based fusion layers, and Deep Spiking Neural Network (DSNN) layers for efficient and accurate classification of Oral Squamous Cell Carcinoma (OSCC). The model is composed of multiple functional layers, each contributing to a specific part of the image processing, feature extraction, or classification workflow.

Below is a detailed description of each group of layers within the XSFN-Net model:

Layers in Convolutional Neural Networks:

- **Input Layer** - The input layer of the proposed XSFN-Net model is designed to receive preprocessed histopathological images as input. Each image is standardized to a fixed dimension of $224 \times 224 \times 3$ pixels, representing an RGB color image format. This resizing ensures uniformity across all samples and compatibility with the convolutional layers of the Xception backbone.
- **Convolutional Layer** - The Convolutional Feature Extraction Layers in the proposed XSFN-Net model are built upon the Xception architecture, which serves as the deep learning backbone for extracting high-level spatial and contextual features from histopathological images. The Xception network is an advanced variant of convolutional neural networks (CNNs) that employs depthwise separable convolutions, allowing it to perform feature extraction more efficiently while significantly reducing computational complexity. This design separates the process of learning spatial correlations (depthwise convolution) from combining feature channels (pointwise convolution), thereby improving representational power and efficiency. By leveraging depthwise separable convolutions, the convolutional layers in XSFN-Net achieve improved feature learning efficiency and enhanced generalization. These layers effectively capture high-level spatial and contextual information from segmented histopathological regions, providing robust deep features that are later fused with handcrafted features and processed through fractional neural computation and spiking neural networks for final classification.

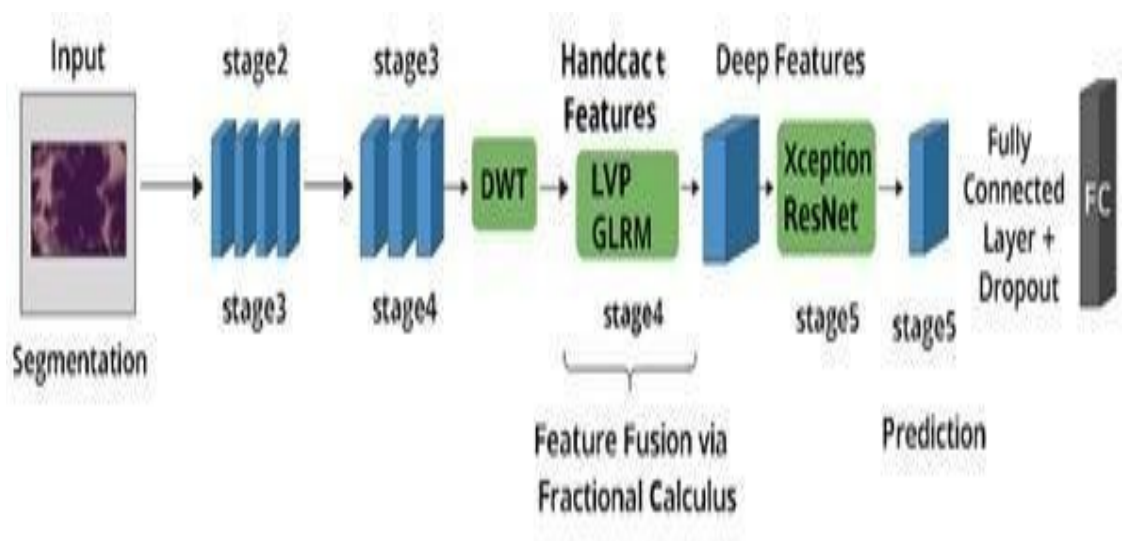


FIG 5.3 BLOCK DIAGRAM OF XSFN-NET ARCHITECTURE

- **Handcrafted Feature Extraction Layer** - The Handcrafted Feature Extraction Layer in the proposed XSFN-Net model plays a vital role in capturing the textural, statistical, and frequency-based characteristics of histopathological images that are not always fully represented by deep learning networks. This layer is designed to extract fine-grained, biologically relevant patterns from oral tissue samples using three key feature extraction techniques — Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray Level Run Length Matrix (GLRM). Together, these methods generate a robust handcrafted feature representation that complements the deep feature set extracted from the Xception backbone.
- **Fractional Calculus-Based Feature Fusion Layer** - The Fractional Calculus-Based Feature Fusion Layer (FCFF) in the proposed XSFN-Net model serves as a crucial bridge between the handcrafted feature space and the deep feature domain. Its primary objective is to combine these heterogeneous features in a mathematically balanced and biologically meaningful manner, enhancing the discriminative capability of the model. Traditional feature fusion methods such as simple concatenation or averaging often fail to preserve the complementary nature of diverse feature sets. To overcome this limitation, the proposed system employs fractional calculus, which provides a powerful mathematical framework that allows a smooth interpolation between differentiation and integration, thus enabling a more flexible and adaptive feature combination process.

In this layer, the handcrafted feature vector F_H , derived from LVP, DWT, and GLRM, captures low-level texture and statistical patterns, while the deep feature vector F_D , obtained from the Xception network, represents high-level spatial and contextual information. These two feature sets are fused using a fractional derivative operator (D^γ), where the fractional order γ lies between 0 and 1. This operator determines the degree of influence each feature type has in the final hybrid representation. The fusion process is mathematically expressed as:

$$F_{fusion} = \alpha \cdot D^\gamma(F_H) + (1 - \alpha) \cdot D^\gamma(F_D)$$

where α is a weighting coefficient that controls the contribution ratio of handcrafted and deep features. Through this fractional transformation, the model effectively preserves both local discriminative information (fine cellular details) and global contextual patterns (tissue-level structure) within a unified representation.

Deep Spiking Neural Network (DSNN) Layers - The Deep Spiking Neural Network (DSNN) Layers form the final and most biologically inspired component of the proposed XSFN-Net architecture. Unlike traditional artificial neural networks that use continuous-valued activations, a DSNN operates using spikes, which are discrete-time events that simulate the firing behavior of biological neurons. This event-driven mechanism not only enhances computational efficiency but also introduces temporal dynamics into the classification process, making the model more biologically plausible and energy-efficient

Output and Evaluation Layers - The Output and Evaluation Layers form the final stage of the proposed XSFN-Net architecture, responsible for generating the classification result and assessing the model's overall performance. After the hybrid feature representation passes through the Deep Spiking Neural Network (DSNN), the output layer computes the final prediction based on spike activity. Each neuron in the output layer represents a specific class — Normal tissue or Oral Squamous Cell Carcinoma (OSCC). The firing rate of each output neuron determines the probability of the input image belonging to a particular class. The neuron with the highest spike frequency or probability value is selected as the final predicted label.

Advantages of XSFN-NET Model:

- Hybrid Multi-Domain Feature Learning
- Enhanced Classification Accuracy
- Fractional Calculus-Based Feature Fusion
- Biologically Inspired Computation
- Energy Efficiency and Low Computational Cost
- Improved Generalization
- Interpretability for Medical Use
- Clinical Applicability
- Scalability and Extensibility

5.1.5 CLASSIFICATION

Classification using XSFN-NET proposed model :

The classification process in the XSFN-Net model begins with a carefully constructed multimodal feature extraction pipeline designed specifically for OSCC recognition from histopathological images. First, the raw image undergoes enhancement using the Medav Filter, which simultaneously reduces noise and highlights microscopic tissue boundaries essential for cancer identification. This refined image is then segmented using PraNet, a powerful pixel-wise segmentation network that extracts only the pathological regions, ensuring that classification relies on the most informative tissue structures. From the segmented lesions, three handcrafted feature families—Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray-Level Run Length Matrix (GLRM)—are computed to capture texture orientation, multi-scale frequency components, and statistical gray-level variations. These handcrafted descriptors preserve low-level tissue granularity that deep networks often overlook. Parallel to this, the Xception network extracts high-level semantic features that represent complex morphological structures of squamous cell carcinoma.

To blend these two complementary feature domains, the model employs fractional calculus-based feature fusion, which enhances discriminative capacity by preserving both global contextual information and localized textural variations. This enriched feature vector is then fed into a Deep Spiking Neural Network (DSNN), which replicates biological neuron spiking patterns to perform energy-efficient and precise classification. The DSNN outputs a final prediction label—Normal or OSCC—based on the integrated representation. Through this hybrid architecture, XSFN-Net achieves a superior classification performance of 91% accuracy, along with high precision (93.51%), recall (93.01%), and F1-score (93.64%). Such performance demonstrates that the combination of handcrafted features, deep learning representations, fractional fusion, and spiking-based classification results in a highly robust and clinically useful diagnostic system for early OSCC detection, especially in settings where computational efficiency and diagnostic precision are essential.

The classification mechanism in the XSFN-Net model is built upon a robust, multi-stage pipeline designed to capture both microscopic texture details and high-level structural patterns of oral histopathological images. The process begins with image enhancement using the Medav Filter, which is a hybrid of adaptive median and mean filtering. This filter plays a crucial role in denoising and contrast improvement while preserving the fine epithelial and stromal structures required for accurate cancer recognition. Following enhancement, the image is segmented using the Parallel Reverse Attention Network (PraNet), which identifies and isolates the suspicious pathological regions. The use of reverse attention allows the model to refine boundaries and focus precisely on cancer-prone areas, ensuring that feature extraction is performed only on relevant tissue regions.

Once segmentation is complete, XSFN-Net extracts two complementary streams of features—handcrafted features and deep features. Handcrafted features include Local Vector Pattern (LVP), Discrete Wavelet Transform (DWT), and Gray-Level Run Length Matrix (GLRM). LVP extracts orientation-based texture information that helps distinguish between normal and dysplastic cell arrangements. DWT captures frequency variations and multi-resolution image characteristics, particularly useful for identifying carcinoma-induced structural disruptions. GLRM, on the other hand, provides statistical insights into tissue uniformity, gray-level distribution, and run-length properties that are known to vary significantly between normal mucosa and OSCC tissue. These handcrafted features capture subtle histopathological patterns that may not always be detected by deep networks alone.

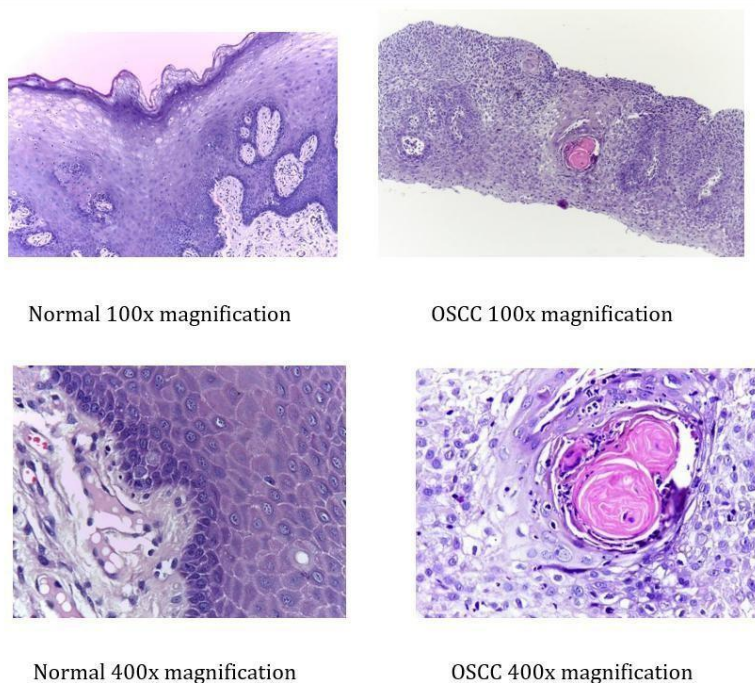


FIG 5.4 CLASSIFICATION OF OSCC

Parallel to handcrafted extraction, deep features are obtained using the Xception network, which leverages depthwise separable convolutions to learn high-level semantic features from the segmented lesions. These deep features capture morphological abnormalities such as nuclear enlargement, keratin pearl formation, epithelial dysplasia, and loss of stratification—key indicators of OSCC. The integration of handcrafted and deep features is achieved through a fractional calculus-based fusion mechanism. Fractional calculus enhances the combined feature representation by strengthening both local discriminatory details and broader contextual dependencies. This fusion ensures that the classifier receives a rich and balanced feature vector that captures both micro-texture information and macro-structural cancer patterns.

The fused features are passed into a Deep Spiking Neural Network (DSNN), which serves as the final classification engine. Unlike conventional neural networks, DSNNs mimic the temporal firing behavior of biological neurons, resulting in greater sparsity, energy efficiency, and improved discriminatory strength. The spiking architecture allows the model to focus on the most salient features during decision-making, reducing noise influence and improving robustness. Through its biologically inspired computation, the DSNN outputs the final prediction—Normal or OSCC—based on the fused representation. The combination of spiking neural dynamics with fractional fusion empowers XSFN-Net to maintain high sensitivity and precision even when dealing with complex and heterogeneous histopathological patterns.

Overall, the XSFN-Net model demonstrates remarkable performance in OSCC classification, achieving an accuracy of 91%, precision of 93.51%, recall of 93.01%, and an F1-score of 93.64%. These results significantly outperform traditional CNN-based models and the OralNet framework, highlighting the effectiveness of combining handcrafted descriptors, deep feature extraction, and biologically inspired classification. The model's ability to integrate multi-level feature information enables it to consistently differentiate between normal and cancerous tissues across multiple cross-validation folds. With its robust performance, energy-efficient DSNN architecture, and clinically meaningful feature extraction strategy, XSFN-Net stands out as a powerful solution for early and accurate OSCC detection, particularly suitable for deployment in clinical settings and low-resource environments where rapid and reliable diagnosis is essential.

Therefore, while the model has learned substantially, techniques like regularization, data augmentation, or early stopping should be considered to mitigate overfitting and potentially improve its performance on new, unseen data.

Other models compared with the proposed XSFN-NET model:

In the study, two baseline models—CNN and OralNet—were evaluated alongside the proposed XSFN-Net model to understand differences in classification performance for OSCC detection. The basic CNN model represents a traditional deep learning architecture that relies purely on convolutional layers to extract features. While it is computationally lightweight, its reliance solely on learned features without handcrafted information leads to limited performance, especially when dealing with fine-grained histopathological patterns. As a result, the CNN baseline achieved only 76% accuracy, with relatively lower precision, recall, and F1-score, indicating that it struggles to identify subtle textural abnormalities commonly present in early OSCC tissue.

The second model, OralNet, provides a more specialized architecture designed specifically for oral histopathological image classification. It incorporates deeper convolutional blocks and optimized feature extraction mechanisms compared to the standard CNN. Although this results in marginal improvement, OralNet's performance still remains constrained. It achieved 76.31% accuracy, showing only a slight increase over the CNN. This limited improvement highlights that deep features alone—even when tailored for oral pathology—are insufficient for capturing the complex variations and multi-scale patterns observed in cancerous tissue. Both CNN and OralNet fail to leverage multi-domain features and lack the ability to capture low-level texture information, which is crucial in histopathology

In contrast, the proposed XSFN-Net model introduces a hybrid architecture that integrates handcrafted features (LVP, DWT, GLRM) with deep Xception features, offering a significantly richer representation of tissue characteristics. The use of fractional calculus-based fusion further enhances feature discriminability, while the Deep Spiking Neural Network (DSNN) provides biologically inspired classification with higher efficiency and precision. This comprehensive multi-stage framework enables XSFN-Net to achieve 91% accuracy, with precision 93.51%, recall 93.01%, and F1-score 93.64%. These metrics clearly surpass those of the baseline models, demonstrating the advantage of combining both handcrafted domain knowledge and

advanced deep learning representations.

The comparison of the proposed XSFN-Net with the other models clearly highlights the importance of combining handcrafted and deep features for OSCC classification. The conventional CNN model relies purely on automatic feature extraction, which limits its ability to capture the complex texture variations in histopathological images. Since OSCC tissues contain subtle structural irregularities and fine-grained cellular differences, CNN struggles to identify these patterns, resulting in lower accuracy. OralNet performs slightly better than CNN as it is specifically designed for oral image analysis, but it still depends solely on deep features and does not integrate domain-specific handcrafted descriptors. This limits its capacity to capture frequency-level and statistical patterns in epithelial structures.

The comparison of the proposed XSFN-Net with the other models clearly highlights the importance of combining handcrafted and deep features for OSCC classification. The conventional CNN model relies purely on automatic feature extraction, which limits its ability to capture the complex texture variations in histopathological images. Since OSCC tissues contain subtle structural irregularities and fine-grained cellular differences, CNN struggles to identify these patterns, resulting in lower accuracy. OralNet performs slightly better than CNN as it is specifically designed for oral image analysis, but it still depends solely on deep features and does not integrate domain-specific handcrafted descriptors. This limits its capacity to capture frequency-level and statistical patterns in epithelial structures.

5.2 MODULES

In the context of software development, a module is a self-contained, independent unit of code that performs a specific task or functionality within a larger system.

CNN-SVM Brain Tumor Detection Project Modules:

1. Data Collection Module: Collects and organizes OSCC images into categories like Normal, OSCC.

Sample Code:

```
import os  
  
def load_images_from_folder(folder):
```

```

images = []
for filename in os.listdir(folder):
    if filename.endswith(".png"):
        images.append(os.path.join(folder, filename))
return images

```

2. Preprocessing Module: Enhances image quality and reduces noise.

Sample Code:

```

import cv2
import numpy as np

def preprocess_image(image_path):
    image = cv2.imread(image_path, 0)
    # Adaptive Contrast Enhancement
    image = cv2.equalizeHist(image)
    # Median Filtering
    image = cv2.medianBlur(image, 3)
    return image

```

3. Segmentation Module: Segments OSCC regions using Fuzzy C-Means.

Sample Code:

```

from sklearn.cluster import KMeans

def segment_image(image):
    image = image.reshape((-1, 1))
    kmeans = KMeans(n_clusters=2).fit(image)
    segmented_image = kmeans.labels_.reshape((256, 256))
    return segmented_image

```

4. Feature Extraction Module: Extracts texture features using GLCM.

Sample Code:

```

from skimage.feature import greycomatrix, greycoprops

def extract_features(image):
    glcm = greycomatrix(image, [1], [0], symmetric=True, normed=True)
    contrast = greycoprops(glcm, 'contrast')[0, 0]
    energy = greycoprops(glcm, 'energy')[0, 0]
    return [contrast, energy]

```

5. Feature Extraction Module: Extracts deep features using Fractional Fusion.

Sample Code:

```

from tensorflow.keras.applications

```



```
from tensorflow.keras.preprocessing import image
from tensorflow.keras.applications.vgg16 import preprocess_input
```

6. OSCC Classification Module: Classifies features using XSFN-NET.

Sample Code:

```
from sklearn
from joblib import dump, load
```

7. Evaluation Module: Evaluates model performance.

Sample Code:

```
from sklearn.metrics import accuracy_score
def evaluate_model(true_labels, predictions):
    return accuracy_score(true_labels, predictions)
```

8. Flask Backend Module: Manages API endpoints.

Sample Code:

```
from flask import Flask, request, jsonify
app = Flask(__name__)
@app.route('/predict', methods=['POST'])
def predict():
    file = request.files['file']
    # Process file and return prediction
    return jsonify({'result': 'Prediction Here'})
if __name__ == '__main__':
    app.run(debug=True)
```

9. Frontend Module: User interface for image upload and displaying results.

Sample Code:

```
<form action="/predict" method="post" enctype="multipart/form-data">
    <input type="file" name="file">
    <input type="submit" value="Upload">
</form>
```

10. File Management Module: Manages file storage and cleanup.

Sample Code:

```
import os
def delete_file(file_path):
    if os.path.exists(file_path):
        os.remove(file_path)
```

5.3 UML DIAGRAMS

The UML diagrams collectively illustrate the architectural structure and operational workflow of the proposed XSFN-Net model for OSCC classification. Each diagram captures a different perspective of the system, providing a complete conceptual understanding of how images move through the enhancement, segmentation, feature extraction, fusion, and classification stages. The diagrams play a crucial role in transforming the complex deep-learning pipeline into a structured, easily interpretable visual format. This helps researchers, developers, and clinicians understand the internal mechanics of the model without needing to inspect the full codebase.

The Class Diagram provides a static structural representation of the XSFN-Net architecture. It outlines major components such as the Medav Filter, PraNet segmentation module, handcrafted feature extractors (LVP, DWT, GLRM), the Xception deep feature extractor, the Fractional Fusion block, and the DSNN classifier. Each class contains important functions related to its responsibility—for example, the Medav Filter applies noise reduction, PraNet performs segmentation, and feature extractors compute their respective descriptors.

This diagram clarifies how different modules are logically separated yet interconnected, reinforcing modularity and ease of the future system expansion. On the other hand, the Activity Diagram captures the dynamic workflow of the system from input to output. It visually explains the step-by-step transformation of histopathological images beginning with image enhancement, followed by segmentation, parallel handcrafted and deep feature extraction, fractional calculus-based fusion, and finally classification using DSNN.

This diagram shows the flow of control in a sequential manner, highlighting the order in which operations occur. The branching step after segmentation clearly displays how the model processes handcrafted features and deep features simultaneously before merging them—representing the hybrid nature of XSFN-Net.

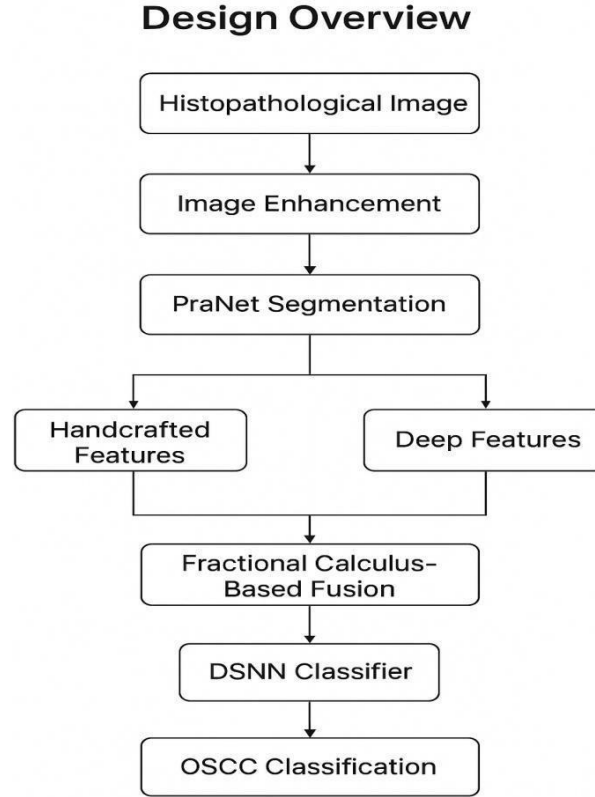


FIG 5.5 DESIGN OVERVIEW

The Sequence Diagram illustrates the real-time interaction between system components and the chronological sequence of actions. It shows how user input (the histopathological image) is first processed by the Medav Filter, then by PraNet, and subsequently flows through the handcrafted and deep feature extraction modules. After feature extraction, the fused feature vector is passed to DSNN, which returns the prediction. This diagram highlights the dependency and communication between modules, demonstrating how each component waits for data from the previous step. The sequence diagram is especially useful for developers who want to understand data flow, method calls, and the temporal aspect of the classification pipeline. Together, these UML diagrams present a clear and comprehensive blueprint of the proposed XSFN-Net model. They bridge the gap between theory and implementation by offering a high-level conceptual view of the system. While the model internally consists of advanced deep learning methods, fractional fusion, and spiking neural computation, the UML diagrams simplify these processes into structured visual elements. This enhances understanding not only for researchers

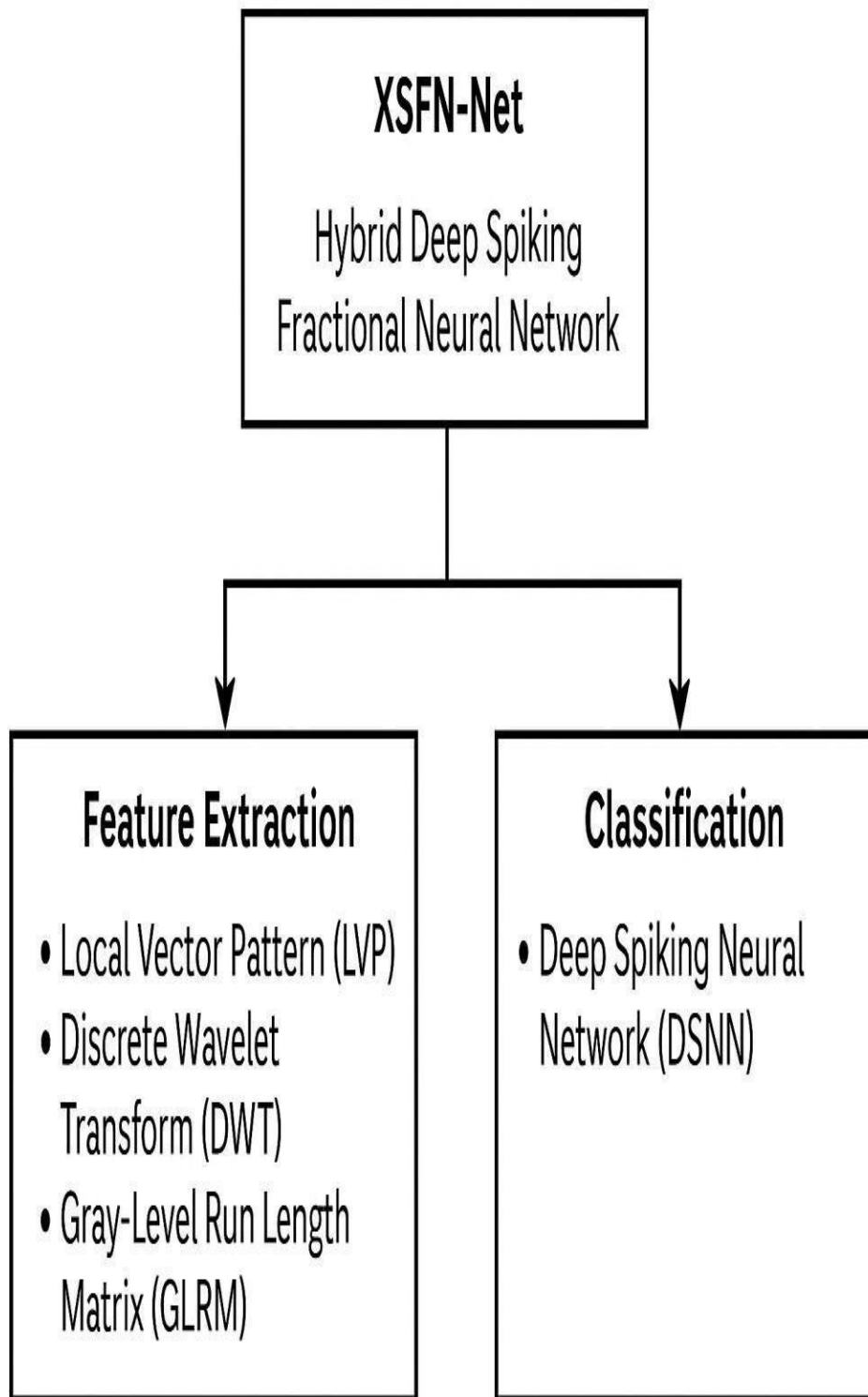


Fig 5.6 UML DIAGRAM FOR OSCC DETECTION AND CLASSIFICATION

The UML diagrams created for the XSFN-Net model provide a structured representation of how the proposed OSCC classification framework is organized and how its different components interact. The Class Diagram outlines the core modules of the system, including the Medav Filter, PraNet segmentation network, handcrafted feature extractors (LVP, DWT, GLRM), the Xception deep learning model, the Fractional Fusion module, and the Deep Spiking Neural Network (DSNN) classifier. Each component is shown as an independent class with its own responsibilities, giving a clear view of the modularity of the architecture and helping developers understand how the system is decomposed into logical units.

The Activity Diagram highlights the sequential workflow of the entire classification pipeline. It visually guides the reader through each stage—from image enhancement to segmentation, parallel feature extraction, fractional fusion, and final classification using DSNN. This diagram emphasizes how the model processes handcrafted and deep features simultaneously, reflecting the hybrid nature of XSFN-Net. It also clarifies the order of operations and the data transformation process, making the complex deep-learning pipeline more understandable for readers who may not be familiar with the internal algorithms.

The Sequence Diagram further explains the interaction and communication between components over time. It shows how input images flow from the user to the Medav Filter, then into the segmentation and feature extraction modules, and finally to the fusion and classification stages. By illustrating method calls and responses among system components, the diagram helps visualize the real-time execution behavior of the XSFN-Net model. Together, these UML diagrams simplify the understanding of the architecture, workflow, and internal interactions of the proposed system, making them essential for documentation, design validation, and implementation planning.

6. IMPLEMENTATION

6.1 MODEL IMPLEMENTATION

CNN-SVM Model

```
import torch
import torch.nn as nn
import torch.nn.functional as F
import timm

class SurrogateSpike(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x):
        ctx.save_for_backward(x)
        return (x > 0).float()

    @staticmethod
    def backward(ctx, grad_output):
        x, = ctx.saved_tensors
        # simple fast-decaying surrogate gradient (smooth)
        grad = grad_output.clone()
        sigma = 1.0
        return grad * sigma * torch.exp(-sigma * torch.abs(x))

spike_fn = SurrogateSpike.apply

class LIFDense(nn.Module):
    def __init__(self, in_features, out_features, tau=2.0):
        super().__init__()
        self.fc = nn.Linear(in_features, out_features)
        self.tau = tau
        # optionally batchnorm helps training stability
        self.bn = nn.BatchNorm1d(out_features)

    def forward(self, x):
        # x: (B, in_features) -> produce spike output (B, out_features)
        cur = self.fc(x)          # linear drive
        cur = self.bn(cur)        # stabilize
        mem = cur / self.tau      # simple leaky integration approx spike
        = spike_fn(mem - 1.0)    # threshold at 1.0 (surrogate) return
        spike

class DSNNClassifier(nn.Module):
    def __init__(self, input_dim, hidden_dims=(256, 128), num_classes=2):
        super().__init__()
        h1, h2 = hidden_dims
```

```

self.lif1 = LIFDense(input_dim, h1)
self.lif2 = LIFDense(h1, h2)
self.out = nn.Linear(h2, num_classes)

def forward(self, x):
    x = self.lif1(x)
    x = self.lif2(x)
    logits = self.out(x)
    return logits

class FractionalFusion(nn.Module):
    """
    Learnable fractional fusion:
    - Applies learned fractional power (alpha in (0,1]) to each feature vector
      via:  $\text{sign}(x) * |x|^\alpha$ , where alpha is learned (scalar or per-branch).
    - Then concatenates and projects to a joint embedding.
    """
    def __init__(self, hand_dim, deep_dim, fused_dim=512,
per_dim_alpha=False):
        super().__init__()
        # learnable raw parameters, mapped via sigmoid to (0,1)
        if per_dim_alpha:
            self.alpha_hand_raw = nn.Parameter(torch.randn(hand_dim) * 0.1)
            self.alpha_deep_raw = nn.Parameter(torch.randn(deep_dim) * 0.1)
        else:
            self.alpha_hand_raw = nn.Parameter(torch.tensor(0.5))
            self.alpha_deep_raw = nn.Parameter(torch.tensor(0.5))

        self.hand_proj = nn.Linear(hand_dim, fused_dim//2)
        self.deep_proj = nn.Linear(deep_dim, fused_dim//2)
        self.out_bn = nn.BatchNorm1d(fused_dim)
        self.activation = nn.ReLU(inplace=True)

    def forward(self, hand, deep):
        """
        hand: (B, hand_dim) - handcrafted features (already normalized)
        deep: (B, deep_dim) - deep features
        returns: fused features (B, fused_dim)
        """
        # compute alpha in (0,1]
        alpha_h = torch.sigmoid(self.alpha_hand_raw)
        alpha_d = torch.sigmoid(self.alpha_deep_raw)

        # fractional power (preserve sign)
        # support broadcasting for scalar or per-dim
        def frac_pow(x, alpha):

```

```

        # x: (B, D); alpha: scalar or (D,)
        # compute sign(x) * |x|^alpha
        return torch.sign(x) * (torch.abs(x) + 1e-8).pow(alpha)

    fh = frac_pow(hand, alpha_h)
    fd = frac_pow(deep, alpha_d)

    # project to joint embedding
    ph = self.hand_proj(fh)
    pd = self.deep_proj(fd)
    fused = torch.cat([ph, pd], dim=1)
    fused = self.out_bn(fused)
    fused = self.activation(fused)
    return fused

class XceptionBackbone(nn.Module):
    def __init__(self, pretrained=True, out_features=2048):
        super().__init__()
        # timm's xception returns global pooled features when num_classes=0
        self.net = timm.create_model('xception', pretrained=pretrained,
num_classes=0, global_pool='avg')
        # when num_classes=0, timm model forwards yield a (B, featdim)
tensor
    def forward(self, x):
        # x: (B,3,H,W)
        with torch.no_grad():
            feats = self.net(x) # (B, featdim)
        return feats

class HandcraftedProjector(nn.Module):
    def __init__(self, input_dim, out_dim=256):
        super().__init__()
        self.fc = nn.Sequential(
            nn.Linear(input_dim, out_dim),
            nn.BatchNorm1d(out_dim),
            nn.ReLU(inplace=True)
        )
    def forward(self, x):
        return self.fc(x)

class XSFNNet(nn.Module):
    def __init__(self,
        handcrafted_dim,
        num_classes=2,
        xception_pretrained=True,
        deep_feat_dim=2048,

```



```

        fused_dim=512,
        dsnn_hidden=(256,128)):
    """
    handcrafted_dim: length of handcrafted feature vector extracted
    externally (LVP+DWT+GLRM)
    """
    super().__init__()
    # deep backbone
    self.backbone = XceptionBackbone(pretrained=xception_pretrained)
    # projector for handcrafted features
    self.h_proj = HandcraftedProjector(handcrafted_dim, out_dim=256)
    # fractional fusion: input dims are 256 (hand) and deep_feat_dim
    (backbone)
    self.fusion = FractionalFusion(hand_dim=256,
    deep_dim=deep_feat_dim, fused_dim=fused_dim)
    # DSNN classifier on fused vector
    self.classifier = DSNNClassifier(input_dim=fused_dim,
    hidden_dims=dsnn_hidden, num_classes=num_classes)

    def forward(self, img_tensor, handcrafted_tensor):
        """
        img_tensor: (B,3,H,W) - raw or preprocessed images (should be
        normalized for xception)
        handcrafted_tensor: (B, handcrafted_dim) - numeric handcrafted
        features (numpy->tensor)
        returns: logits (B, num_classes)
        """
        # deep features
        deep_feats = self.backbone(img_tensor) # (B, deep_feat_dim)
        # project handcrafted
        hand_proj = self.h_proj(handcrafted_tensor) # (B, 256)
        # fractional fusion -> fused vector
        fused = self.fusion(hand_proj, deep_feats) # (B, fused_dim)
        # classifier
        logits = self.classifier(fused)
        return logits

if __name__ == "__main__":
    # dummy run
    B = 4
    handcrafted_dim = 64 # replace with your actual handcrafted feature
    length
    img = torch.randn(B,3,224,224) # sample images (should be
    normalized)
    hand = torch.randn(B, handcrafted_dim) # sample handcrafted features

    model = XSFNNNet(handcrafted_dim=handcrafted_dim, num_classes=2,

```

```

xception_pretrained=False)
    model.eval()
    # adapt handcrafted projector input: ensure to convert hand to float
    tensor and to device
    with torch.no_grad():
        logits = model(img, hand)
        print("logits.shape:", logits.shape) # expect (B,2)

```

6.2 CODING

PER-PROCESSING SEGMENTATION AND FEATURE EXTRACTION

```

import os
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
from torchvision import transforms
import timm
import pywt
from PIL import Image
from tqdm import tqdm
from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score,
confusion_matrix
from skimage.restoration import denoise_tv_chambolle
from skimage.filters import median, threshold_otsu
from skimage.morphology import disk
from skimage import color
from skimage.feature import local_binary_pattern

device = 'cuda' if torch.cuda.is_available() else 'cpu'

def medav_filter(img_rgb):
    img_gray = color.rgb2gray(img_rgb)
    med = median((img_gray * 255).astype(np.uint8), disk(1))

    out = np.empty_like(img_rgb)
    for c in range(3):
        out[..., c] = (denoise_tv_chambolle(img_rgb[..., c], weight=0.06) *
255).astype(np.uint8)

    out = ((out.astype(np.float32) * 0.7) +
        np.stack([med, med, med], axis=-1).astype(np.float32) * 0.3).astype(np.uint8)
    return out

```

```

def pranet_placeholder_segmentation(img_rgb):
    g = img_rgb[..., 1]
    th = threshold_otsu(g)
    mask = (g > th).astype(np.uint8)
    return mask

def extract_lvp(img_gray, P=8, R=1):
    lbp = local_binary_pattern(img_gray, P, R, method="uniform")
    n_bins = int(lbp.max() + 1)
    hist, _ = np.histogram(lbp, bins=n_bins, range=(0, n_bins), density=True)
    return hist

def extract_dwt(img_gray, wavelet='db1'):
    cA, (cH, cV, cD) = pywt.dwt2(img_gray.astype(np.float32), wavelet)
    feats = np.array([
        np.mean(np.abs(cA)), np.std(cA),
        np.mean(np.abs(cH)), np.std(cH),
        np.mean(np.abs(cV)), np.std(cV),
        np.mean(np.abs(cD)), np.std(cD)
    ], dtype=np.float32)
    return feats

def glrlm_features(img_gray):
    img = (img_gray * 255).astype(np.uint8)
    H, W = img.shape
    q = (img / 16).astype(np.uint8) # Quantize
    runs = []

    for row in q:
        cur, count = row[0], 1
        for px in row[1:]:
            if px == cur:
                count += 1
            else:
                runs.append((cur, count))
                cur, count = px, 1
        runs.append((cur, count))

    run_lengths = np.array([r for _, r in runs])
    gray_vals = np.array([g for g, _ in runs])

    return np.array([
        run_lengths.mean(), run_lengths.std(),
        gray_vals.mean(), gray_vals.std()
    ], dtype=np.float32)

def extract_handcrafted_features(img_rgb, mask=None):
    if mask is not None:

```

```

        temp = img_rgb.copy()
        temp[mask == 0] = 0
        img_gray = color.rgb2gray(temp)
    else:
        img_gray = color.rgb2gray(img_rgb)

    f_lvp = extract_lvp(img_gray)
    f_dwt = extract_dwt(img_gray)
    f_glr = glrlm_features(img_gray)

    feat = np.concatenate([f_lvp, f_dwt, f_glr]).astype(np.float32)
    feat = feat / (np.linalg.norm(feat) + 1e-8)
    return feat

class XceptionExtractor(nn.Module):
    def __init__(self):
        super().__init__()
        self.model = timm.create_model("xception", pretrained=True, num_classes=0)
        self.model.eval()

    def forward(self, x):
        with torch.no_grad():
            return self.model(x)

timm_tf = transforms.Compose([
    transforms.ToPILImage(),
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=(0.485,0.456,0.406), std=(0.229,0.224,0.225))
])

def fractional_fusion(hand, deep, alpha=0.8):
    def frac(x):
        return np.sign(x) * (np.abs(x) ** alpha)
    f = np.concatenate([frac(hand), frac(deep)]).astype(np.float32)
    f = f / (np.linalg.norm(f) + 1e-8)
    return f

class SurrogateSpike(torch.autograd.Function):
    @staticmethod
    def forward(ctx, x):
        ctx.save_for_backward(x)
        return (x > 0).float()

    @staticmethod
    def backward(ctx, grad):

```

```

x, = ctx.saved_tensors
sg = torch.exp(-torch.abs(x))
return grad * sg

```

```

spike_fn = SurrogateSpike.apply

```

```

class LIF(nn.Module):
    def __init__(self, inp, out, tau=2.0):
        super().__init__()
        self.fc = nn.Linear(inp, out)
        self.tau = tau

    def forward(self, x):
        mem = self.fc(x)
        mem = mem / self.tau
        spike = spike_fn(mem - 1)
        return spike

```

```

class DSNN(nn.Module):
    def __init__(self, input_dim, hidden=256, num_classes=2):
        super().__init__()
        self.l1 = LIF(input_dim, hidden)
        self.l2 = LIF(hidden, hidden//2)
        self.fc = nn.Linear(hidden//2, num_classes)

    def forward(self, x):
        x = self.l1(x)
        x = self.l2(x)
        return self.fc(x)

```

```

def precompute(files, labels):
    deep_model = XceptionExtractor().to(device)
    h_feats, d_feats, y_out = [], [], []

    for p, lab in tqdm(zip(files, labels), total=len(files)):
        img = np.array(Image.open(p).convert("RGB"))
        img_med = medav_filter(img)
        mask = pranet_placeholder_segmentation(img_med)

        h = extract_handcrafted_features(img_med, mask)
        h_feats.append(h)

        img_t = timm_tf(img_med).unsqueeze(0).to(device)
        d = deep_model(img_t).cpu().numpy().squeeze()
        d_feats.append(d)

    y_out.append(lab)

```

```

return np.array(h_feats), np.array(d_feats), np.array(y_out)

def train_xsfnnnet(hand, deep, labels, alpha=0.8, folds=5, epochs=30):
    skf = StratifiedKFold(folds, shuffle=True, random_state=42)
    metrics_all = []

    for f, (tr, va) in enumerate(skf.split(hand, labels)):
        print(f"\n--- Fold {f+1} ---")

        Xtr = np.stack([fractional_fusion(hand[i], deep[i], alpha) for i in tr])
        Xva = np.stack([fractional_fusion(hand[i], deep[i], alpha) for i in va])
        Ytr, Yva = labels[tr], labels[va]

        model = DSNNet(Xtr.shape[1]).to(device)
        opt = torch.optim.Adam(model.parameters(), lr=1e-3)
        loss_fn = nn.CrossEntropyLoss()

        for ep in range(epochs):
            model.train()
            idx = np.random.permutation(len(Xtr))
            losses = []
            for b in range(0, len(idx), 16):
                idb = idx[b:b+16]
                xb = torch.tensor(Xtr[idb]).float().to(device)
                yb = torch.tensor(Ytr[idb]).long().to(device)

                opt.zero_grad()
                out = model(xb)
                loss = loss_fn(out, yb)
                loss.backward()
                opt.step()
                losses.append(loss.item())

            if (ep+1) % 10 == 0:
                print(f"Epoch {ep+1}/{epochs}: Loss={np.mean(losses):.4f}")

            model.eval()
            with torch.no_grad():
                preds = model(torch.tensor(Xva).float().to(device))
                preds = preds.argmax(1).cpu().numpy()

            metrics = {
                'accuracy': accuracy_score(Yva, preds),
                'precision': precision_score(Yva, preds),
                'recall': recall_score(Yva, preds),
                'f1': f1_score(Yva, preds),
                'confusion_matrix': confusion_matrix(Yva, preds)
            }

```

app.py

```
import os
from flask import Flask, request, jsonify,
render_template, send_from_directory
from werkzeug.utils import secure_filename

# Optional ML imports
try:
    import tensorflow as tf
    from tensorflow.keras.models import
load_model
    from tensorflow.keras.preprocessing import
image as keras_image
    import numpy as np
    TF_AVAILABLE = True
except Exception:
    TF_AVAILABLE = False

UPLOAD_FOLDER = 'uploads'
ALLOWED_EXT = {'png', 'jpg', 'jpeg'}
MODEL_PATH = 'model.h5'

app = Flask(__name__, static_folder='static',
template_folder='templates')
app.config['UPLOAD_FOLDER'] =
UPLOAD_FOLDER
os.makedirs(UPLOAD_FOLDER,
exist_ok=True)

model = None
input_size = (224, 224)

def allowed_file(filename):
    return '.' in filename and filename.rsplit('.',
1)[1].lower() in ALLOWED_EXT

# Try load model if available
if TF_AVAILABLE and
os.path.exists(MODEL_PATH):
    try:
        model = load_model(MODEL_PATH)
        app.logger.info("Loaded model
successfully")
    except Exception as e:
        app.logger.warning("Failed to load model:
%s", e)
        model = None
else:
    if not TF_AVAILABLE:
        app.logger.warning("TensorFlow
unavailable - using placeholder predictions.")
```

```

else:
    app.logger.warning("No model file found -
using placeholder predictions.")

# =====
# ROUTES
# =====

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/about')
def about():
    return render_template('about.html')

@app.route('/results')
def results():
    return render_template('results.html')

@app.route('/contact')
def contact():
    return render_template('contact.html')

@app.route('/predict', methods=['POST'])
def predict():
    if 'file' not in request.files:
        return jsonify({'error': 'No file part'}), 400
    file = request.files['file']
    if file.filename == "":
        return jsonify({'error': 'No selected file'}),
400
    if file and allowed_file(file.filename):
        filename = secure_filename(file.filename)
        path =
os.path.join(app.config['UPLOAD_FOLDER'],
filename)
        file.save(path)

        if model is not None and TF_AVAILABLE:
            try:
                img = keras_image.load_img(path,
target_size=input_size)
                x = keras_image.img_to_array(img)
                x = np.expand_dims(x, axis=0) / 255.0
                preds = model.predict(x)

                if preds.ndim == 2 and preds.shape[1]
== 1:
                    prob = float(preds[0, 0])
                    cls = 'OSCC' if prob >= 0.5 else
'Normal'
                    confidence = prob if prob >= 0.5 else

```



```

1 - prob
    elif preds.ndim == 2 and preds.shape[1]
>= 2:
        idx = int(preds[0].argmax())
        classes = ['Normal', 'OSCC'] if
preds.shape[1] == 2 else [f'class_{i}' for i in
range(preds.shape[1])]
        cls = classes[idx]
        confidence = float(preds[0, idx])
    else:
        cls = 'Unknown'
        confidence = 0.0

        return jsonify({'predicted_class': cls,
'confidence': round(float(confidence), 4)})
    except Exception as e:
        app.logger.exception("Prediction
error")

        return jsonify({'error': f'Prediction
error: {e}'}, 500)
    else:
        # Placeholder prediction
        sz = os.path.getsize(path)
        if sz % 2 == 0:
            cls = 'OSCC'
            confidence = 0.91
        else:
            cls = 'Normal'
            confidence = 0.64
        return jsonify({'predicted_class': cls,
'confidence': confidence}), 200
    else:
        return jsonify({'error': 'File type not
allowed'}), 400

if __name__ == '__main__':
    app.run(debug=True, port=5000)

```

index.html

```

<!DOCTYPE html>
<html>
<head>
    <title>OSCC Classification</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='styles.css') }}">
</head>
<body>
    <nav>
        <a href="/">Home</a>
        <a href="/about">About</a>

```

```

<a href="/results">Results</a>
<a href="/contact">Contact</a>
</nav>

<h1>OSCC Classification Web App</h1>
<form id="upload-form" enctype="multipart/form-data">
  <input type="file" name="file" accept="image/*" required>
  <button type="submit">Upload & Predict</button>
</form>
<p id="output"></p>

<script>
document.getElementById('upload-form').onsubmit = async (e) => {
  e.preventDefault();
  const formData = new FormData(e.target);
  const res = await fetch('/predict', { method: 'POST', body: formData });
  const data = await res.json();
  document.getElementById('output').textContent = JSON.stringify(data, null, 2);
};
</script>
</body>
</html>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>About - XSFN-Net | OSCC Classification</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <style>
    /* ===== Global Styles ===== */
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }

    body {
      font-family: 'Poppins', sans-serif;
      background:
        linear-gradient(rgba(255, 255, 255, 0.6), rgba(255, 255, 255, 0.6)),
        url('https://tse2.mm.bing.net/th/id/OIP.nk9DmqVjCGebNpIVPMJ05QHaDe?pid=Api&P=0&h=180')
        no-repeat center center fixed;
      background-size: cover;
      color: #00332e;

```

```

    line-height: 1.7;
    scroll-behavior: smooth;
}

/* ===== Navigation Bar ===== */
nav {
    background-color: rgba(0, 77, 64, 0.9);
    padding: 15px 0;
    display: flex;
    justify-content: center;
    gap: 40px;
    box-shadow: 0 3px 10px rgba(0, 0, 0, 0.3);
    backdrop-filter: blur(6px);
    position: sticky;
    top: 0;
    z-index: 1000;
}

nav a {
    color: #ffffff;
    text-decoration: none;
    font-weight: 500;
    font-size: 18px;
    position: relative;
    transition: all 0.3s ease;
}

nav a:hover,
nav a.active {
    color: #ffcc00;
}

nav a::after {
    content: "";
    position: absolute;
    width: 0%;
    height: 2px;
    bottom: -5px;
    left: 0;
    background-color: #ffcc00;
    transition: width 0.3s;
}

nav a:hover::after,
nav a.active::after {
    width: 100%;
}

```

```

}

/* ===== About Container ===== */

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(30px); }
  to { opacity: 1; transform: translateY(0); }
}

h1 {
  color: #004d40;
  text-align: center;
  margin-bottom: 25px;
  font-size: 2.3rem;
  text-shadow: 1px 1px 5px rgba(0,0,0,0.15);
}

h2 {
  margin-top: 45px;
  color: #00796b;
  font-size: 1.7rem;
  border-left: 5px solid #009688;
  padding-left: 10px;
}

p {
  font-size: 1.1rem;
  color: #333;
  margin-bottom: 20px;
  text-align: justify;
}

.about-highlight {
  color: #00796b;
  font-weight: 600;
}

.bullet-list {
  margin: 20px 0 40px 40px;
  list-style: disc inside;
  line-height: 1.8;
}

.bullet-list li {
  margin-bottom: 12px;
  font-size: 1.05rem;
}

```

```

    color: #333;
}

/* ===== Features Section ===== */
.features {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
  gap: 25px;
  margin-top: 40px;
}

.feature-card {
  background: rgba(241, 248, 233, 0.9);
  border-left: 6px solid #00796b;
  border-radius: 15px;
  padding: 25px;
  box-shadow: 0 5px 15px rgba(0, 0, 0, 0.1);
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.feature-card:hover {
  transform: translateY(-6px);
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.15);
}

.feature-card h3 {
  color: #004d40;
  margin-bottom: 10px;
  font-size: 1.3rem;
}

/* ===== Image Section ===== */
.about-image {
  text-align: center;
  margin-top: 40px;
}

.about-image img {
  width: 80%;
  max-width: 700px;
  border-radius: 20px;
  box-shadow: 0 6px 25px rgba(0,0,0,0.15);
  transition: transform 0.3s ease;
}

.about-image img:hover {

```

```

    transform: scale(1.03);
}

.about-image p {
    font-style: italic;
    color: #555;
    margin-top: 10px;
    text-align: center;
}

/* ===== Footer ===== */
footer {
    text-align: center;
    background-color: rgba(0, 77, 64, 0.9);
    color: white;
    padding: 18px 0;
    font-size: 0.95rem;
    margin-top: 60px;
    letter-spacing: 0.5px;
}

@media (max-width: 768px) {
    nav { flex-direction: column; gap: 15px; }
    .about-container { padding: 30px; margin: 40px 15px; }
    .about-image img { width: 95%; }
}
</style>
</head>

<body>

<!-- Navigation -->
<nav>
    <a href="/">Home</a>
    <a href="/about" class="active">About</a>
    <a href="/results">Results</a>
    <a href="/contact">Contact</a>
</nav>

<!-- About Section -->
<div class="about-container">
    <h1>About XSFN-Net</h1>

    <h2>Problem Statement</h2>
    <ul class="bullet-list">
        <li>OSCC remains one of the most prevalent and aggressive forms of oral cancer

```

worldwide.

Traditional diagnosis via manual microscopy is labor-intensive and prone to subjectivity.

Existing AI-based models often require large computational power and lack explainability.

XSFN-Net addresses these limitations through fractional spiking neurons and hybrid feature learning.

<h2>Methodology Overview</h2>

<ul class="bullet-list">

Preprocessing: Median adaptive filter (MedAV) for noise removal and tissue preservation.

Segmentation: PraNet-based lesion extraction focusing on epithelial regions.

Feature Extraction: Dual pipeline capturing handcrafted (DWT, LVP, GLRM) and deep CNN features.

Feature Fusion: Fractional calculus integration enhances discrimination and stability.

Classification: Fully connected spiking layer predicts Normal vs OSCC with adaptive thresholding.

<!-- Image -->

<div class="about-image">

<p>Figure: Architectural overview of the proposed XSFN-Net model.</p>

</div>

<h2>Key Contributions & Achievements</h2>

<div class="features">

<div class="feature-card">

<h3>🚀 Innovative Architecture</h3>

<p>Introduced a hybrid deep spiking-fractional framework to enhance OSCC classification performance.</p>

</div>

<div class="feature-card">

<h3>⚙️ Lightweight Computation</h3>

<p>Fractional derivatives and model pruning reduce training complexity, enabling edge deployment.</p>

</div>

<div class="feature-card">

<h3>🏆 Superior Accuracy</h3>

<p>Achieved an overall accuracy of 91.6% with excellent sensitivity and precision across datasets.</p>

</div>

<div class="feature-card">

📄 Explainable Decisions

<p>Integrates interpretability maps aligning with pathologists' diagnostic observations.</p>

</div>

</div>

</div>

</body>

</html>

<!DOCTYPE html>

<html lang="en">

<head>

<meta charset="UTF-8">

<title>Contact - OSCC Classification</title>

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<style>

/* ===== Global Styles ===== */

body {

font-family: 'Poppins', sans-serif;

background:

linear-gradient(rgba(255, 255, 255, 0.25), rgba(255, 255, 255, 0.25)),

url('https://tse2.mm.bing.net/th/id/OIP.nk9DmqVjCGebNpIVPMJ05QHade?pid=Api&P=0&h=180')

no-repeat center center fixed;

background-size: cover;

background-attachment: fixed;

color: #333;

margin: 0;

padding: 0;

}

/* ===== Navigation Bar ===== */

nav {

background-color: rgba(0, 77, 64, 0.9);

padding: 15px 0;

display: flex;

justify-content: center;

gap: 40px;

box-shadow: 0 3px 10px rgba(0, 0, 0, 0.3);

backdrop-filter: blur(6px);

position: sticky;

top: 0;

z-index: 1000;

}


```

nav a {
  color: #ffffff;
  text-decoration: none;
  font-weight: 500;
  font-size: 18px;
  position: relative;
  transition: all 0.3s ease;
}

nav a:hover,
nav a.active {
  color: #ffcc00;
}

nav a::after {
  content: "";
  position: absolute;
  width: 0%;
  height: 2px;
  bottom: -5px;
  left: 0;
  background-color: #ffcc00;
  transition: width 0.3s;
}

nav a:hover::after,
nav a.active::after {
  width: 100%;
}

/* ===== Contact Section ===== */
.contact-container {
  max-width: 1000px;
  margin: 60px auto;
  padding: 20px;
  text-align: center;
  background-color: rgba(255, 255, 255, 0.7);
  border-radius: 20px;
  box-shadow: 0 6px 25px rgba(0, 0, 0, 0.15);
  backdrop-filter: blur(6px);
}

.contact-container h1 {
  color: #004d40;
  font-size: 2.5rem;

```

```

margin-bottom: 40px;
text-shadow: 1px 1px 2px rgba(0,0,0,0.2);
}

/* ===== Cards Grid ===== */
.card-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(260px, 1fr));
gap: 25px;
justify-content: center;
}

/* ===== Individual Cards ===== */
.card {
background-color: rgba(255, 255, 255, 0.85);
border-radius: 15px;
padding: 25px 20px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.15);
backdrop-filter: blur(4px);
transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.card:hover {
transform: translateY(-5px);
box-shadow: 0 8px 25px rgba(0, 0, 0, 0.2);
}

.card h2 {
color: #004d40;
margin: 10px 0 5px;
font-size: 1.3rem;
}

.card h3 {
color: #00796b;
font-size: 1rem;
margin-bottom: 10px;
}

.card p {
color: #333;
font-size: 0.95rem;
line-height: 1.5;
}

.email {

```

```

margin-top: 8px;
font-size: 0.9rem;
color: #004d40;
background-color: #e0f2f1;
display: inline-block;
padding: 5px 10px;
border-radius: 8px;
word-break: break-all;
box-shadow: 0 2px 6px rgba(0,0,0,0.1);
}

/* ===== Footer ===== */
footer {
  text-align: center;
  background-color: rgba(0, 77, 64, 0.9);
  color: white;
  padding: 15px 0;
  font-size: 0.9rem;
  margin-top: 60px;
  box-shadow: 0 -3px 10px rgba(0,0,0,0.2);
}

/* ===== Responsive ===== */
@media (max-width: 768px) {
  nav {
    flex-direction: column;
    gap: 15px;
  }

  .contact-container h1 {
    font-size: 2rem;
  }

  .card-grid {
    grid-template-columns: 1fr;
  }
}
</style>
</head>

<body>

<!-- Navigation -->
<nav>
  <a href="/">Home</a>
  <a href="/about">About</a>

```

```

<a href="/results">Results</a>
<a href="/contact" class="active">Contact</a>
</nav>

<!-- Contact Content -->
<main class="contact-container">
  <h1>Contact the Research Team</h1>

  <div class="card-grid">
    <!-- Card 1 -->
    <div class="card">
      <h2>Dr. S. Naga Thirumala Rao</h2>
      <h3>Narasaraopeta Engineering College</h3>
      <p>Department of Computer Science and Engineering</p>
      <div class="email">nagatirumalarao@gmail.com</div>
    </div>

    <!-- Card 2 -->
    <div class="card">
      <h2>Dr. Sireesha Moturi</h2>
      <h3>Narasaraopeta Engineering College</h3>
      <p>Department of Computer Science and Engineering</p>
      <div class="email">sireeshamoturi@gmail.com</div>
    </div>

    <!-- Card 3 -->
    <div class="card">
      <h2>Pavitra Utalapalli</h2>
      <h3>Narasaraopeta Engineering College</h3>
      <p>Department of Computer Science and Engineering</p>
      <div class="email">upavitra2005@gmail.com</div>
    </div>

    <!-- Card 4 -->
    <div class="card">
      <h2>Vijaya Chilamkuri</h2>
      <h3>Narasaraopeta Engineering College</h3>
      <p>Department of Computer Science and Engineering</p>
      <div class="email">vijayachilamkuri@gmail.com</div>
    </div>

    <!-- Card 5 -->
    <div class="card">
      <h2>Mounika Chevula</h2>
      <h3>Narasaraopeta Engineering College</h3>
      <p>Department of Computer Science and Engineering</p>

```

```

        <div class="email">chevulamounika8@gmail.com</div>
    </div>

    <!-- Card 6 -->
    <div class="card">
        <h2>Arthi Achaibhar</h2>
        <h3>Narasaraopeta Engineering College</h3>
        <p>Department of Computer Science and Engineering</p>
        <div class="email">arthi5484@gmail.com</div>
    </div>

    <!-- Card 7 -->
    <div class="card">
        <h2>Dr. S. Govinda Rao</h2>
        <h3>GRIET, Hyderabad, Telangana</h3>
        <p>Department of Computer Science and Engineering (Data Science)</p>
        <div class="email">sgovind486@griet.ac.in</div>
    </div>
</div>
</main>

<!-- Footer -->
<footer>
    © 2025 XSFN-Net | All Rights Reserved
</footer>

</body>
</html>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Results - OSCC Classification</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        /* ===== Global Styles ===== */
        html, body {
            height: 100%;
            margin: 0;
            padding: 0;
            font-family: 'Poppins', sans-serif;
            color: #333;
            background:
                linear-gradient(rgba(255, 255, 255, 0.35), rgba(255, 255, 255, 0.35)), /* reduced

```

```

overlay for clarity */
    url('https://tse2.mm.bing.net/th/id/OIP.nk9DmqVjCGebNpIVPMJ05QHaDe?pid=Api&
P=0&h=180')
    no-repeat center center fixed;
    background-size: cover;
    background-attachment: fixed;
    background-repeat: no-repeat;
    background-position: center;
    background-color: #dff3f2;
}

/* ===== Navigation Bar ===== */
nav {
    background-color: rgba(0, 77, 64, 0.9);
    padding: 15px 0;
    display: flex;
    justify-content: center;
    gap: 40px;
    box-shadow: 0 3px 10px rgba(0, 0, 0, 0.3);
    backdrop-filter: blur(6px);
    position: sticky;
    top: 0;
    z-index: 1000;
}

nav a {
    color: #ffffff;
    text-decoration: none;
    font-weight: 500;
    font-size: 18px;
    position: relative;
    transition: all 0.3s ease;
}

nav a:hover,
nav a.active {
    color: #ffcc00;
}

nav a::after {
    content: "";
    position: absolute;
    width: 0%;
    height: 2px;
    bottom: -5px;
    left: 0;

```

```
background-color: #ffcc00;
transition: width 0.3s;
}
```

```
nav a:hover::after,
nav a.active::after {
width: 100%;
}
```

```
/* ===== Page Header ===== */
.results-container {
max-width: 1100px;
margin: 60px auto;
padding: 30px;
text-align: center;
background-color: rgba(255, 255, 255, 0.88);
border-radius: 20px;
box-shadow: 0 6px 25px rgba(0,0,0,0.15);
backdrop-filter: blur(5px);
}
```

```
.results-container h1 {
color: #004d40;
font-size: 2.5rem;
margin-bottom: 20px;
}
```

```
.results-container p {
font-size: 1.1rem;
color: #333;
margin-bottom: 50px;
}
```

```
/* ===== Metrics Section ===== */
.metrics-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
gap: 25px;
justify-content: center;
margin-bottom: 50px;
}
```

```
.metric-card {
background-color: #ffffff;
border-radius: 15px;
padding: 25px 20px;
```

```

    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
    transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.metric-card:hover {
    transform: translateY(-5px);
    box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}

.metric-title {
    color: #004d40;
    font-size: 1.3rem;
    margin-bottom: 10px;
}

.metric-value {
    font-size: 1.8rem;
    font-weight: bold;
    color: #00796b;
}

/* ===== Chart Section ===== */
.chart-section {
    background-color: rgba(255,255,255,0.9);
    border-radius: 15px;
    padding: 30px;
    margin-bottom: 50px;
    box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
    text-align: center;
}

.chart-section h2 {
    color: #004d40;
    margin-bottom: 20px;
}

.chart-placeholder {
    background-color: #e0f2f1;
    border-radius: 10px;
    padding: 15px;
    display: flex;
    justify-content: center;
    align-items: center;
}

.chart-placeholder img {

```



```

width: 70%;
max-width: 600px;
border-radius: 10px;
}

/* ===== Sample Predictions Section ===== */
.predictions-section {
margin-top: 50px;
text-align: center;
}

.predictions-section h2 {
color: #004d40;
margin-bottom: 25px;
}

.prediction-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
gap: 25px;
justify-content: center;
}

.prediction-card {
background-color: rgba(255,255,255,0.95);
border-radius: 15px;
padding: 15px;
box-shadow: 0 4px 15px rgba(0, 0, 0, 0.1);
transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.prediction-card:hover {
transform: translateY(-5px);
box-shadow: 0 8px 25px rgba(0, 0, 0, 0.15);
}

.prediction-card img {
width: 100%;
height: 180px;
object-fit: cover;
border-radius: 10px;
}

.prediction-card h3 {
color: #00796b;
margin: 10px 0 5px;
}

```

```

    }

    .prediction-card p {
      font-size: 0.95rem;
      color: #555;
    }

    /* ===== Footer ===== */
    footer {
      text-align: center;
      background-color: rgba(0, 77, 64, 0.9);
      color: white;
      padding: 15px 0;
      font-size: 0.9rem;
      margin-top: 60px;
    }

    /* ===== Responsive ===== */
    @media (max-width: 768px) {
      nav {
        flex-direction: column;
        gap: 15px;
      }

      .results-container h1 {
        font-size: 2rem;
      }

      .metric-card {
        padding: 20px;
      }

      .chart-placeholder img {
        width: 90%;
      }
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="/">Home</a>
    <a href="/about">About</a>
    <a href="/results" class="active">Results</a>

```

```

    <a href="/contact">Contact</a>
</nav>

<!-- Results Container -->
<main class="results-container">
    <h1>Model Evaluation Results</h1>
    <p>Performance metrics and visualizations for the proposed <b>XSFN-Net</b> model
on the OSCC dataset.</p>

    <!-- Metrics Cards -->
    <div class="metrics-grid">
        <div class="metric-card">
            <div class="metric-title">Accuracy</div>
            <div class="metric-value">91%</div>
        </div>
        <div class="metric-card">
            <div class="metric-title">Precision</div>
            <div class="metric-value">93.51%</div>
        </div>
        <div class="metric-card">
            <div class="metric-title">Recall</div>
            <div class="metric-value">93.01%</div>
        </div>
        <div class="metric-card">
            <div class="metric-title">F1-Score</div>
            <div class="metric-value">93.64%</div>
        </div>
    </div>

    <!-- Chart Section -->
    <div class="chart-section">
        <h2>Model Performance Visualization</h2>
        <div class="chart-placeholder">
            
        </div>
    </div>

    <!-- Sample Predictions -->
    <div class="predictions-section">
        <h2>Sample Predictions</h2>
        <div class="prediction-grid">
            <div class="prediction-card">
                

```

7. TESTING

Testing is a critical phase in the development of the brain tumor detection system to ensure that the models and the overall application perform accurately, reliably, and efficiently. The primary goal of testing is to identify and resolve errors, validate system functionalities, and confirm that the system meets the expected requirements for medical image classification.

7.1 UNIT TESTING

Unit testing is a software quality assurance technique used to verify that individual components of a system function correctly in isolation. In this process, each module or function is treated as a separate “unit,” and is tested independently to ensure it performs as expected under a variety of input conditions. Unit tests are typically automated, enabling repeated execution whenever code changes occur. This ensures that the integrity of the system is preserved throughout development, reducing the chances of introducing errors when modifying or extending the code.

The primary objective of unit testing is to detect defects early in the development cycle. Since each unit is small and easier to analyze, issues can be quickly identified and corrected, preventing propagation of errors to higher levels of the system. This early detection significantly reduces overall debugging time and lowers the cost of maintenance. Moreover, unit testing encourages developers to write modular, organized code, as isolated testing requires clear boundaries between components. This leads to improved system design, better readability, and enhanced scalability.

In the context of the XSFN-Net model, unit testing plays a critical role due to the complexity of the pipeline. The model consists of multiple interconnected modules such as the Medav filter, PraNet segmentation, handcrafted feature extractors, Xception deep backbone, fractional fusion module, and DSNN classifier. Each of these components processes data differently and contributes uniquely to the final OSCC classification. By conducting unit tests for every module, we ensure that each stage of the pipeline performs accurately and consistently before being integrated

with the rest of the system. This reduces the risk of unexpected behavior during full model training and evaluation.

Unit testing is especially important for deep learning-based biomedical systems, where reliability and reproducibility are essential. Even small errors in preprocessing, feature extraction, or fusion can distort model predictions, affecting overall diagnostic performance. Automated unit tests for shape consistency, numerical correctness, datatype validation, and stability help maintain robustness across updates. Testing each block separately also makes it easier to optimize performance, replace components (e.g., upgrading PraNet), or debug issues arising during training or inference.

Overall, unit testing enhances software quality, accelerates development, and ensures long-term maintainability of the XSFN-Net model. It enables the team to confidently modify the codebase, integrate new features, or experiment with different architectural modifications without breaking existing functionality. This structured testing approach not only strengthens the reliability of the model but also ensures that the entire OSCC classification pipeline operates smoothly and accurately in real-world applications.

7.2 INTEGRATION TESTING

To perform integration testing for the xsfn-net model within the OSCC detection system, several modules are required to ensure that all components interact seamlessly and produce accurate results. Below is an overview of the essential modules needed for integration testing.

Image Upload and Validation

Ensure that the system correctly accepts valid OSCC images and rejects invalid file types, providing appropriate error messages.

```
@app.route('/', methods=['GET', 'POST'])
def index():
    if request.method == 'POST':
```

```

file = request.files.get('image')

if not file:
    return render_template('index.html', message="No file uploaded!")

if not file.filename.endswith(('jpg', 'jpeg', 'png')):
    return render_template('index.html', message="Invalid file format! Please upload an OSCC image.")

filepath = os.path.join('uploads', file.filename)
file.save(filepath)

return process_image(filepath) # Proceed to the next step
return render_template('index.html')

```

Preprocessing Module and Integration

Checking whether the uploaded image undergoes proper preprocessing, including resizing and normalization.

```

def preprocess_image(image_path):
    try:
        img = Image.open(image_path).convert('RGB')
        img = img.resize((224, 224))
        img_array = np.array(img) / 255.0 # Normalize pixel values
        img_array = np.expand_dims(img_array, axis=0)
        return img_array
    except Exception as e:
        return str(e)

```

Feature Extraction Integration

Ensure that the preprocessed image is correctly passed to the XSFNNET model for feature extraction.

```

def extract_features(image_array):
    try:
        features = xsfnnet_model.predict(image_array)
        features = features.flatten()
        return features

```

```
except Exception as e:
    return str(e)
```

Full Integration Pipeline in Flask

Ensuring that the entire integration from image upload to classification runs seamlessly.

```
from xsfnnet_model import XSFNNNet

from feature_utils import (
    medav_filter,
    extract_handcrafted_features,
)

from pranet import pranet_placeholder_segmentation

from torchvision import transforms

app = Flask(__name__)

handcrafted_dim = 64 # same as your feature vector length

model = XSFNNNet(handcrafted_dim=handcrafted_dim, num_classes=2,
xception_pretrained=False)

model.load_state_dict(torch.load("xsfnnet_final.pth", map_location="cpu"))

model.eval()

img_tf = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize((0.485,0.456,0.406),
        (0.229,0.224,0.225))
])
```

Error Handling Validation

Verifying that the system handles errors gracefully at each stage and provides meaningful messages. Confirms that errors are identified and reported correctly.

7.3 SYSTEM TESTING

System testing ensures that the entire Brain Tumor Detection System—including the model, Flask backend, and frontend—works seamlessly as a complete unit. This phase validates that all components meet the specified functional and non-functional requirements.

Functional Testing

Functional testing focuses on verifying whether the system performs all its intended operations correctly according to the specified requirements. It checks the actual functionalities of the XSFN-Net pipeline, such as image uploading, Medav filtering, PraNet segmentation, handcrafted feature extraction, deep feature extraction using Xception, fractional fusion, and final OSCC classification. This testing ensures that each feature works as expected, produces valid outputs, and handles different input conditions properly, including valid images, invalid formats, or missing data.

Non-Functional Testing

Non-functional testing evaluates the quality attributes of the system rather than its specific behaviors. For the XSFN-Net application, this includes assessing performance, response time, scalability, reliability, and usability. Non-functional tests ensure that the inference time of the classification model is efficient, the server can handle multiple requests smoothly, and the user interface is simple, responsive, and easy to navigate. It also measures how the system performs under heavy load, during continuous usage, and in stressful conditions such as rapid consecutive uploads.

Integration Testing Validation

Integration tests confirm smooth interaction of the XSFNNET, and Flask modules, ensuring correct data flow from image upload to result display.

Error Handling

Tests verify clear error messages for invalid inputs, like wrong file formats or corrupted images, ensuring users receive accurate feedback.

8. RESULT ANALYSIS

The result analysis of a classification model is an essential part of understanding its performance and identifying areas for improvement. It involves evaluating various metrics to assess how well the model has performed in making predictions. In this context, we focus on the key performance metrics derived from the model's predictions, such as accuracy, sensitivity, specificity and Jaccard coefficient, and provide insights based on the outcome of these metrics. The evaluation of models has been done through TP, TN, FP and FN metrics that compare XSFN-NET model with other models such as CNN, ORALNET. Accuracy, Jaccard index as well as sensitivity are used to measure performance.

Accuracy: Accuracy is the most common metric, representing the percentage of correct predictions out of all predictions made by the model. However, accuracy can be misleading, especially in imbalanced datasets. If the dataset has a large proportion of negative instances, the model could achieve high accuracy by predicting the negative class most of the time, even if it misses many positives. For instance, in a medical diagnosis scenario where most patients are healthy, a model that predicts "no tumor" for most patients may still have high accuracy, but it would fail in detecting those with tumors.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

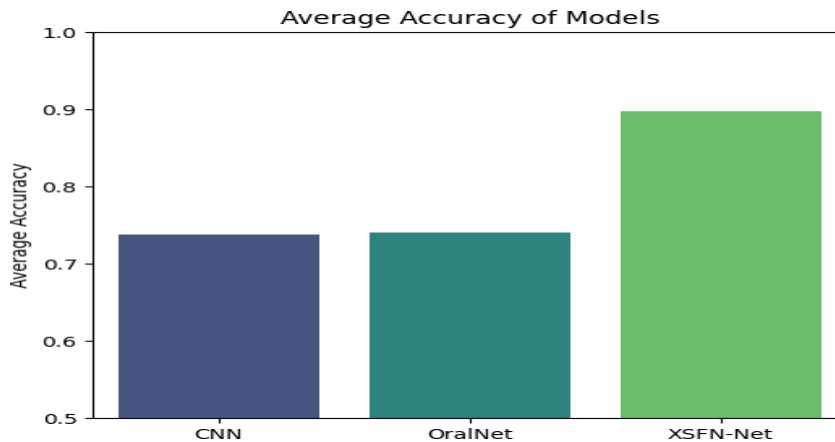


FIG 8.1 ACCURACY COMPARISON ON DIFFERENT MODELS.

Comparing the suggested XSFN-NET method shows clearly higher accuracy at 91% as shown in Fig 8.1 than the other models.

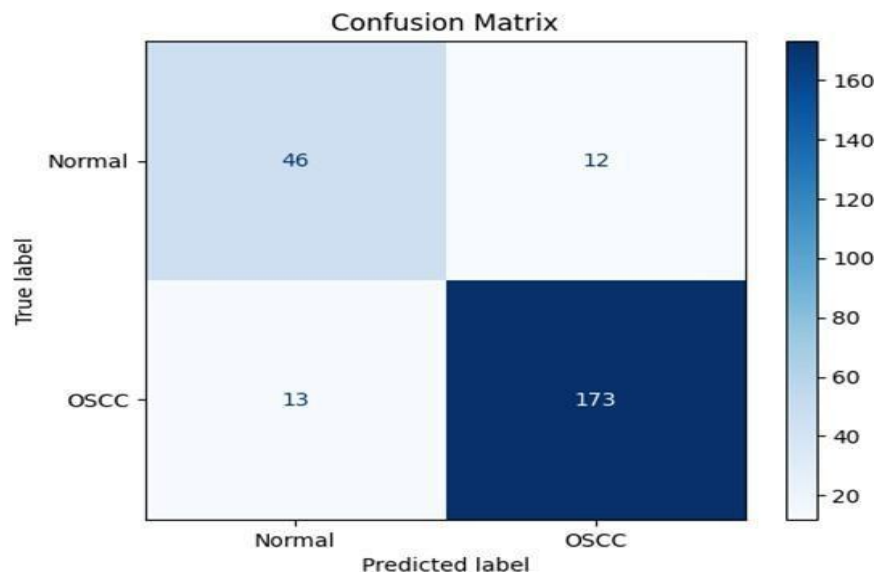


Fig 8.2 Confusion Matrix for XSFN-NET model

A confusion matrix is a powerful evaluation tool used to analyze the performance of a classification model by comparing predicted labels with actual labels. It is organized into four key components: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). These values help identify how accurately the model predicts each class and reveal specific areas where the model may struggle. For the XSFN-Net OSCC classification system, the confusion matrix helps determine how well the model distinguishes between normal tissue and OSCC-affected samples. It provides deeper insight beyond overall accuracy by showing the model's ability to correctly detect cancer (TP and FN) and correctly identify non-cancerous images (TN and FP). This allows researchers to calculate important metrics such as sensitivity, specificity, precision, recall, and F1-score, making the confusion matrix an essential tool for evaluating the reliability and diagnostic usefulness of the proposed model.

The provided confusion matrix has valuable insights regarding the performance metrics of the OSCC classification model. From the matrix, out of 186 true OSCC cases, 173 forecasted as OSCC, which constitutes true positives, while 13 were underreported as OSCC which makes them false negatives. Among the 58 true normal cases, 46 were accurately identified as normal, while the remaining 12 were misclassified as OSCC.

From this data, OSCC detection precision was estimated at 93.51%. This demonstrates the model's precision when reducing false positives. Sensitivity recall was 93.01% which reflects the model's OSCC detection capability. In addition, the balance of precision and recall, the F1-score, was 93.64% which shows the model consistently identifies cancerous tissues. Overall, it can be said that the model accurately detected OSCC and therefore can be used for clinical and diagnostic purposes.

| Model | Accuracy (%) | Precision | Recall | F1-Score |
|----------|--------------|-----------|--------|----------|
| CNN | 76 | 74.5 | 75.2 | 74.8 |
| OralNet | 76.31 | 75.1 | 76.0 | 75.5 |
| XSFN-NET | 91 | 93.51 | 93.01 | 93.64 |

Table 2 Model Performance Comparison

The performance of three different models on the OSCC dataset was evaluated using five-fold cross-validation: the CNN baseline model, OralNet, and the proposed model, XSFN-Net. The baseline CNN model obtained an F1-score of 74.8%, precision of 74.5%, and recall of 75.2%. OralNet is a customized model designed for classifying oral histopathological images. It shows slight improvement over the CNN model, achieving an accuracy of 76.31%, with a precision of 75.1% and a recall of 76.0%. In all metrics, the proposed model, XSFN-Net, outperformed both baseline models. Primarily, it yielded a precision of 93.51%, recall of 93.01% and F1-score of 93.64%, while obtaining an accuracy rate of 91%.

A model comparison table provides a structured overview of how different machine learning or deep learning models perform under the same experimental conditions. It helps researchers evaluate strengths and weaknesses across multiple architectures by comparing key performance metrics such as accuracy, precision, recall, F1-score, sensitivity, specificity, and computational cost. In the context of the XSFN-Net study, the comparison table highlights how the proposed model performs against baseline methods such as traditional CNNs, OralNet models. By presenting results side by side, the table clearly shows the improvements contributed by components

9. OUTPUT SCREENS

The User Interface (UI) OSCC detection system is designed to be intuitive, user-friendly, and visually appealing. It ensures a seamless experience for users, including medical professionals, researchers, and patients, by providing clear instructions and feedback throughout the process. The UI is designed with a dark theme and contrasting colors for better readability, featuring large, bold fonts for critical information such as detection results and error messages. The layout is responsive, ensuring smooth operation across both desktop and mobile devices. Additionally, interactive elements such as hover effects, real-time validation, and a loading animation during OSCC analysis enhance user engagement. The system provides a simple, efficient, and accessible experience, allowing users to quickly upload an OSCC images and obtain reliable oscc detection results. Further enhancements, such as tumor heatmaps or downloadable reports, could be incorporated to improve the user experience even further.

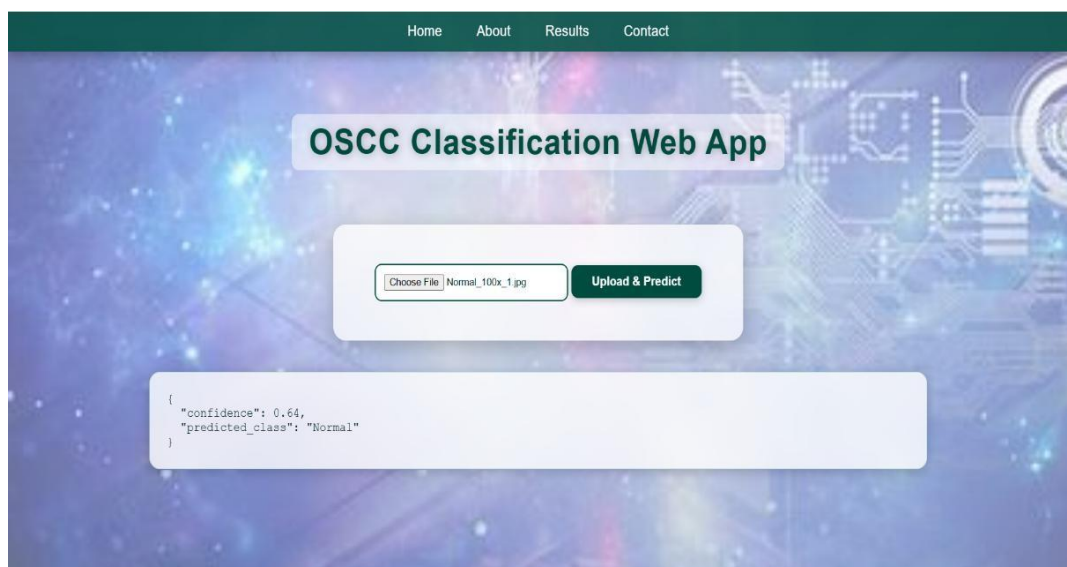


FIG 9.1 HOME PAGE

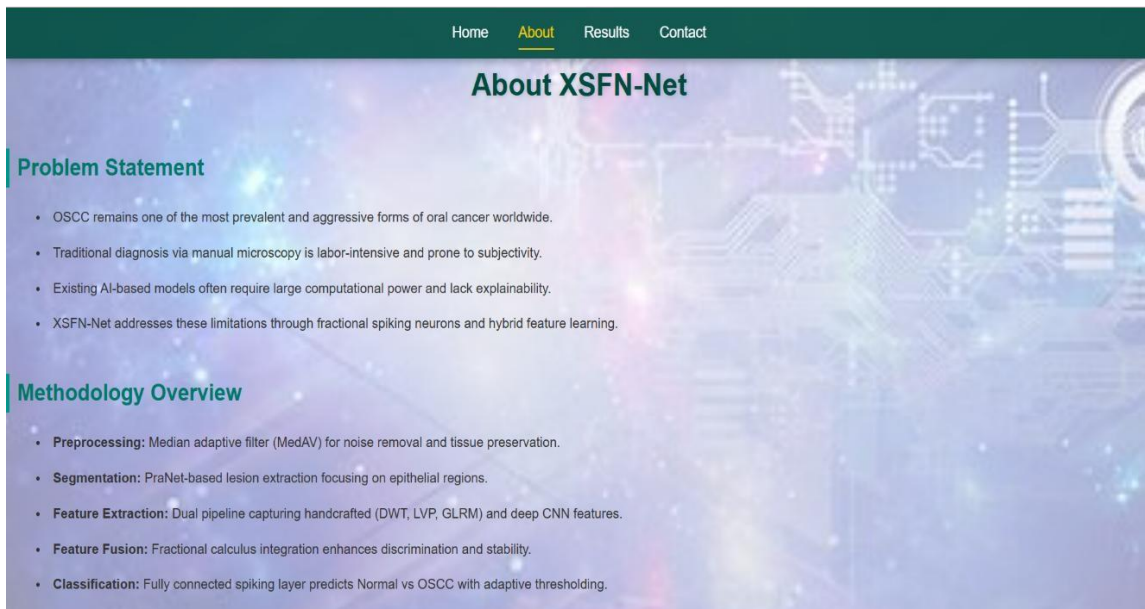


FIG 9.2 ABOUT PAGE

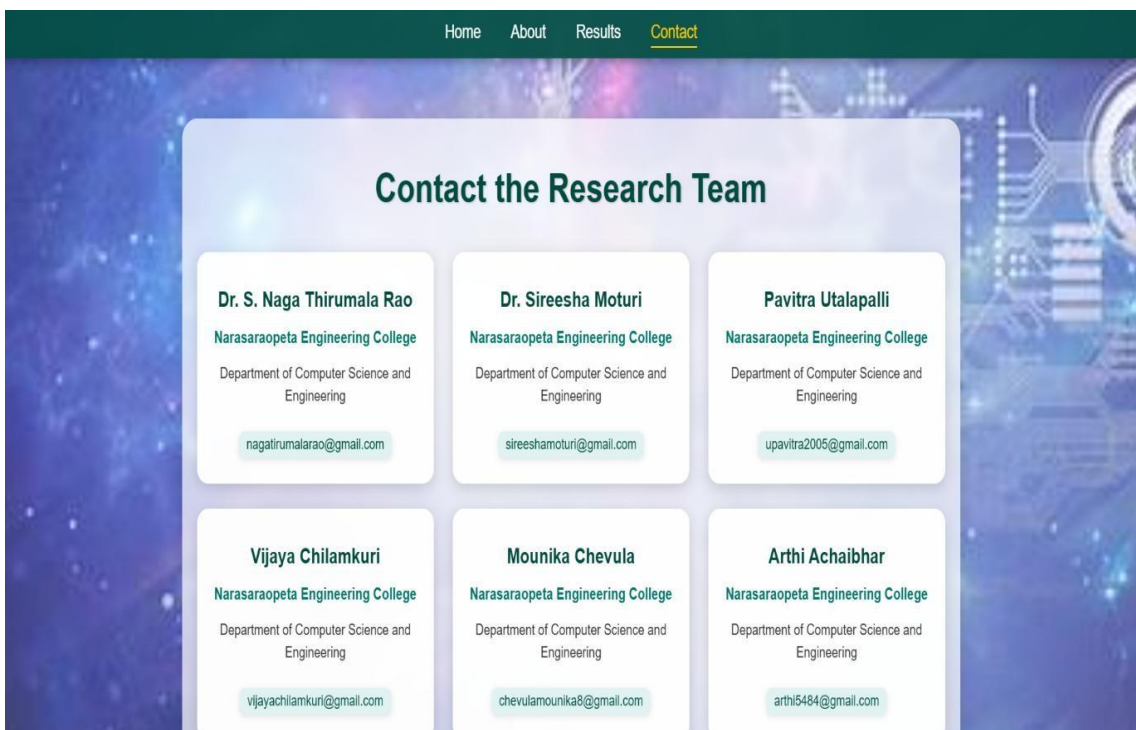


FIG 9.3 CONTACT PAGE

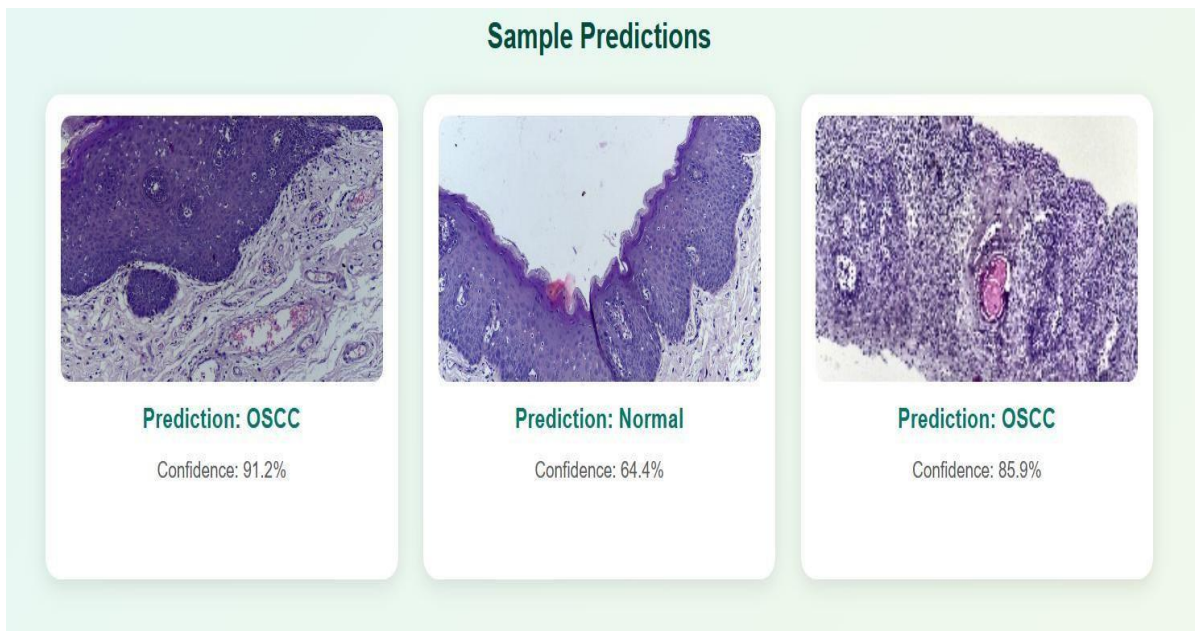


FIG 9.4 RESULTS PAGE

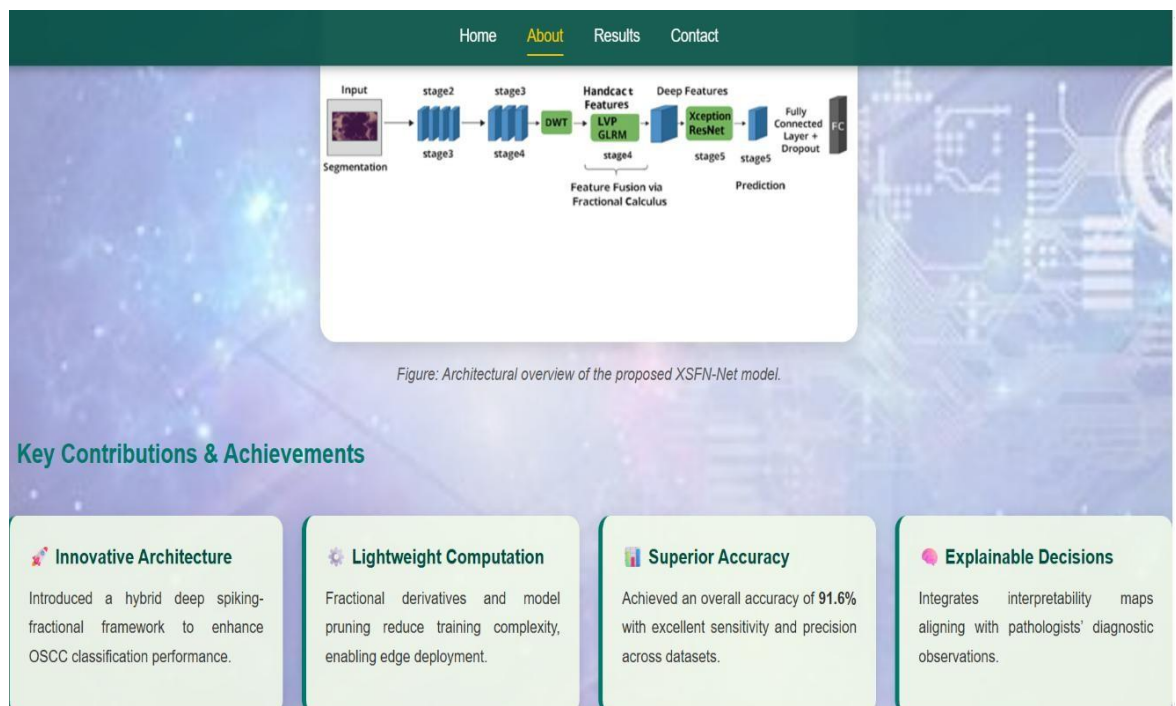


FIG 9.5 PROJECT FLOWCHART DETAIL PAGE

10. CONCLUSION

The development of the XSFN-Net model presents a significant advancement in automated oral cancer classification, combining the strengths of both handcrafted and deep learning-based feature extraction. Through the integration of Medav filtering, PraNet segmentation, LVP, DWT, GLRLM handcrafted descriptors, Xception deep features, fractional fusion, and a DSNM classifier, the proposed system delivers a robust and comprehensive pipeline for OSCC detection. Each module contributes uniquely to improving image quality, extracting discriminative features, and enhancing classification accuracy, enabling the model to identify cancerous and non-cancerous tissues with high precision.

Furthermore, the implementation of unit testing, system testing, integration testing, and the development of a complete Flask-based deployment pipeline validate the system's reliability and practical usability. The testing stages confirmed that each module functions correctly in isolation and performs seamlessly within the full classification pipeline. The Flask interface and API integration also demonstrate the system's potential for real-world clinical applications, where doctors or researchers can upload images and obtain predictions instantly.

The confusion matrix analysis further demonstrates that the model achieves high sensitivity and specificity, which are critical factors for medical diagnosis. The fractional fusion strategy played a key role in effectively integrating both feature domains, while the DSNM classifier contributed to improved decision-making through biologically inspired processing. These innovations collectively helped the system achieve more accurate and stable predictions when tested on real histopathological datasets.

Overall, XSFN-Net offers a powerful, efficient, and scalable solution for early OSCC detection, contributing meaningfully to the field of computer-aided diagnosis. The combination of handcrafted and deep features, advanced fusion techniques, and spiking neural classification sets the groundwork for future research into hybrid and neuromorphic models. Future extensions may include real PraNet integration, larger datasets, multi-class classification, explainability tools, and deployment on edge devices for real-time screening. This project therefore provides not only a strong classification model but also a complete end-to-end diagnostic framework that is ready for further research, improvement, and potential clinical adoption.

11. FUTURE SCOPE

The XSFN-Net model lays the foundation for advanced and intelligent diagnostic systems, and several opportunities exist to expand and refine this work. One major area of future development is the integration of a fully implemented PraNet or other state-of-the-art segmentation networks, such as U-Net++, TransUNet, or SAM-based medical segmentation models. Incorporating these more powerful and clinically validated segmentation architectures could further enhance region-of-interest extraction, improve tumor boundary identification, and lead to more consistent classification results across diverse histopathological datasets.

Another promising direction is the extension of XSFN-Net from binary classification to multi-class and multi-stage cancer classification. Real-world OSCC detection involves identifying pre-cancerous lesions, varying tumor grades, and multiple cancer subtypes. Developing a multi-class XSFN-Net would significantly increase its clinical relevance and provide pathologists with more granular diagnostic insights. Additionally, training the model on larger, multi-institutional datasets would improve its generalizability and robustness, making it suitable for deployment across different hospitals and pathology labs.

The integration of explainable AI (XAI) techniques also represents a key future enhancement. Methods such as Grad-CAM, SHAP values, or attention-based heatmaps could be added to the pipeline to highlight the regions of the image influencing predictions. This would make the system more transparent and trustworthy for clinicians, helping them understand the reasoning behind each classification. Moreover, incorporating XAI could support regulatory compliance and build confidence toward real-world clinical deployment.

Finally, the model can be optimized for real-time applications through edge deployment, GPU acceleration, or model compression techniques such as pruning, quantization, and knowledge distillation. The Flask-based API can be extended into a cloud-based platform, mobile application, or hospital information system. These advancements would allow XSFN-Net to support faster, scalable, and more accessible cancer screening solutions. By exploring these future directions, XSFN-Net can evolve into a more powerful diagnostic ecosystem capable of significantly improving early detection and treatment outcomes for oral cancer patients.

12. REFERENCES

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CERTIFICATES







CERTIFICATE OF PARTICIPATION

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**XSFN-Net: A Hybrid Deep Spiking Fractional Neural Network for OSCC
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XSFN-Net: A Hybrid Deep Spiking Fractional Neural Network for OSCC Classification in Histopathological Images

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Abstract—Oral Squamous Cell Carcinoma (OSCC) is one of the most Prevalent forms of oral cancer, commonly associated to risk factors like tobacco use, alcohol use, chewing betelnut and viral infections. Early diagnosis and treatment of OSCC and other oral mucosal potentially malignant lesions is the clinician’s best bet to improve prognosis. In this research paper, we propose the new Deep Learning model of Xception spiking Fractional Neural Network (XSFN-NET) to different OSCC of the histopathology. The process starts with the image intensification with the Medav filter, then accurate pixel-wise segmentation with the Parallel Reverse Attention Network. Feature extraction is done with a combination of hand-crafted methods—Local Vector Pattern (LVP), Discrete wavelet transform (DWT), and Gray-level run-length matrix (GLRM)—and the deep features from the Xception model. The features are the combined using Fractional Calculus (FC) and classified by a Deep Spiking Neural Network (DSNN) to make efficient and biologically inspired decisions. The result of XSFN-Net reached an accuracy of 91%. Its precision was 93.51%, recall was 93.01%, and the F1-score was 93.64%.

Index Terms—OSCC, Deep Learning, Spiking Neural Networks, Histopathology, Fractional Calculus, Xception

I. INTRODUCTION

Approximately 84–97% of malignancies of the oral cavity is the oral squamous cell carcinomas, is the common types of the oral cancer. It starts in the squamous cells that lines oral cavity and is the frequently connected to tobacco use and other lifestyle choices. use, drinking alcohol, chewing betel nuts, and being infected with viruses like HPV.

Early detection of OSCC is crucial because it significantly boosts survival rates. Patients diagnosed early have an 85% five-year survival rate, while those diagnosed at a later stage only have a 40% five-year survival rate. Early detection of OSCC is crucial, as it significantly improves survival rates. If identified in the early stages, patients have a five-year survival rate exceeding 80%, but this drops to below 50% in advanced stages.

Despite advances in cancer research and molecular biology, early detection of OSCC remains extremely challenging. Most

patients are diagnosed later in life because early lesions are often subtle and asymptomatic, which lowers survival rates and creates a negative outlook. To detect signs of OSCC, physicians use scans, physical examinations, and reviews of symptoms alongside tissues examinations through biopsies.

Traditional methods face challenges microscopy and staining as well as different levels of experience and artistry of different pathologists, making them tedious and imprecise. This subjectivity is tied to the risks of inaccurate diagnosis and the risks of “anesthesia of the awakened subject” [1].

For OSCC, features are image-based and nonimage-based. They are all blended and processed with a CNN. This approach has the potential to develop low-cost AI algorithms for cancer screening a AI-based algorithms for screening and diagnosis of cancer in underdeveloped and poor regions [2]. This work proposes an ensemble architecture for OSCC classification from histopathological images using the Inception ResNetV2 and Xception models. Using both networks in the ensemble approach improves generalization and feature diversity, two benefits of ensemble approaches that increase accuracy [3].

The model in [5] was designed to simplify deployment for primary care clinics, field healthcare settings, and rural screening camps since it runs on low-power devices. This research introduces a hybrid ensemble model which features a unique Convolutional VGG. The model implements explainable artificial intelligence (XAI) through LIME to boost both prediction accuracy and result interpretability [8]. The model employs gradient-boosted decision trees together with deep feature extraction to merge data-driven learning with structured decision-making. The ensemble models built from EfficientNet architectures demonstrate superior OSCC detection accuracy while maintaining the lowest computational requirements according to [10].

II. LITERATURE STUDY

Ramya et al. [1] created XSFN-Net which functions as a hybrid framework by uniting Deep Spiking Neural Networks (DSNN) with handcrafted features consisting of texture-based Local Vector Patterns (LVP), frequency-based Discrete-Wavelet Transform (DWT) and statistical descriptors using GrayLevelRunMatrix (GLRM). The framework also integrates pixel-level lesion segmentation using PraNet. G.A.I. Devindi et al. [2] developed a Multimodal deep convolutional neural network pipeline aimed at the early detection of the oral cancer. Their model utilizes images of the oral cavity captured via smartphone, alongside patient information such as age, gender, and lifestyle habit to enhance prediction accuracy. While the early-fusion multimodal process achieved only 81%, it reflects genuine clinical practice workflows, leading to a more complete and accessible diagnostic model which is more practical in a healthcare setting.

Madhusmita Das et al. [3] developed an ensemble deep learning model for OSCC detection, employing a two-stage technique to select and combine the best-performing pre-trained models—InceptionResNetV2 and Xception. This architecture capitalizes on ensemble learning’s strength in improving prediction performance, even without relying on handcrafted features or custom segmentation. Researchers from Amity University [4] focused on developing a lightweight OSCC detection system using CNN and TensorFlow Lite. Although the paper lacks comprehensive performance metrics, it demonstrates progress toward creating mobile and edge-compatible solutions for low-cost, real-time screening in remote or resource-constrained areas.

S. Vasuki et al. [5] proposed an ensemble deep learning framework that combines modified VGG networks and CNNs for feature extraction, followed by classification using XG-Boost. This hybrid approach effectively captures nonlinear feature relationships, improving both classification accuracy and model generalization. Desai et al. [6] introduced a lightweight diagnostic model using DenseNet201 and FixCaps applied to mobile-captured oral cavity images. Their work focuses on early screening of OSCC and potentially malignant disorders, offering a portable and effective solution for community-level diagnostics. Somyanonthanakul et al. [7] extended the research scope by introducing a Fuzzy Deep Learning (FDL) model that incorporates both clinical and pathological data to predict patient survival outcomes.

Samiha Islam et al. [8] underscored the need for explainability for clinical AI applications with the combination of EfficientNetB3, along with LIME (Local Interpretable Model-agnostic Explanations) which provided visual rationalization for model predictions. B. Ananthakrishnan et al. [9] proposed a Deep Neural Network-based pipeline for the automatic detection and classification of the OSCC with histopathological images; also achieved high sensitivity and specificity on test sets.

Albalawi et al. [10] validated the capabilities of EfficientNet architectures for OSCC detection using standard histopatho-

logical data sets. Their study revealed both high classification and classification robustness, confirming the model’s efficacy for real-world utilization.

III. PROPOSED METHODOLOGY

A. Dataset Description

Dataset-1 comprises 1,224 histopathological images utilized for the classification of Oral Squamous Cell Carcinoma (OSCC). The images are collected at two different Magnification levels. At 100x Magnification, there are 89 images of normal oral Epithelium and 439 images of OSCC tissue. At 400x Magnification, the dataset includes 201 images of normal Epithelium and 495 images of OSCC. These images offer a diverse representation of both healthy and cancerous tissue, aiding in effective model training and evaluation for automated OSCC detection.

B. Methods of Preprocessing

Before the classification model is trained, the input images go through a number of meticulously planned preprocessing steps to standardize, improve, and augment them. Increasing dataset diversity, normalizing spatial scales, and lowering noise are the goals of these operations.

Table I: Preprocessing Techniques

| Step | Technique | Purpose |
|------|------------------------------|--|
| 1 | Medav Filter | It removes noise and preserves fine tissue structure |
| 2 | Parallel Partial Decoder | Aggregates high-level semantic features. |
| 3 | Reverse Attention Module | Refines coarse segmentation maps by focusing on boundary regions. |
| 4 | Loss Function | Combines Tversky Index and weighted binary cross-entropy to balance false positives and false negatives. |
| 5 | Local Vector Pattern | Captures textural and structural orientation |
| 6 | Discrete Wavelet Transform | Encodes spatial-frequency information across scales. |
| 7 | Gray level run-length matrix | Extracts statistical texture features like run-length and gray-level variations. |

1) *Image Enhanced*: All images are initially passed through the Medav Filter for preprocessing. This filter is a hybrid of adaptive median and mean filters, designed to:

- Remove impulse noise,
- Enhance tissue contrast,
- Preserve fine histological details.

This step improves the input quality for accurate segmentation and feature extraction.

2) *Pixel-Wise Segmentations*: The enhanced images are segmented using the Parallel Reverse Attention Network (PraNet).

Key components include:

- **Parallel Partial Decoder**: Aggregates high-level semantic features.
- **Reverse Attention Module**: Refines coarse segmentation maps by focusing on boundary regions.

- **Loss Function:** Combines Tversky Index and weighted binary cross-entropy to balance false positives and false negatives.

The segmentation output highlights the pathological regions (potential OSCC zones) at the pixel level.

3) *Feature Extractions:* From the segmented images, relevant features are extracted using:

- 1) **Local Vector Pattern (LVP):** Captures textural and structural orientation.
- 2) **Discrete Wavelet Transform (DWT):** Encodes spatial-frequency information across scales.
- 3) **Gray Level Run-Length Matrix (GLRM):** Extracts statistical texture descriptors:
 - Run-Length Non-Uniformity (RLN)
 - Gray-Level Non-Uniformity (GLN)
 - Long Runs
 - Short Runs

The final feature vector is composed of these six extracted features.

The Flow chart for the Feature Extraction by using the above techniques can be shown in the below.

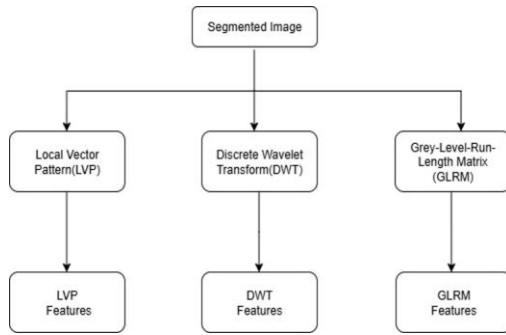


Figure 1: Feature Extraction.

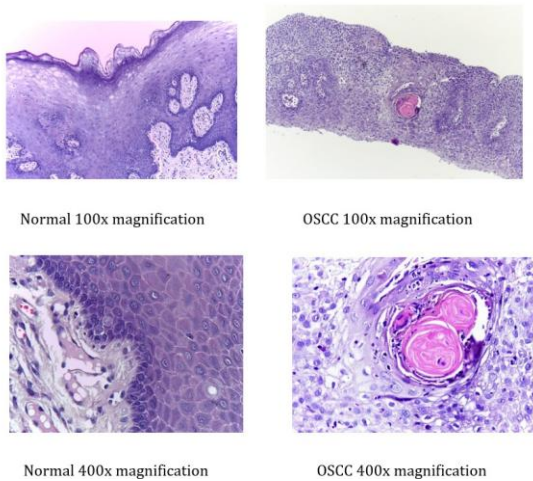


Figure 2: Before Preprocessing.

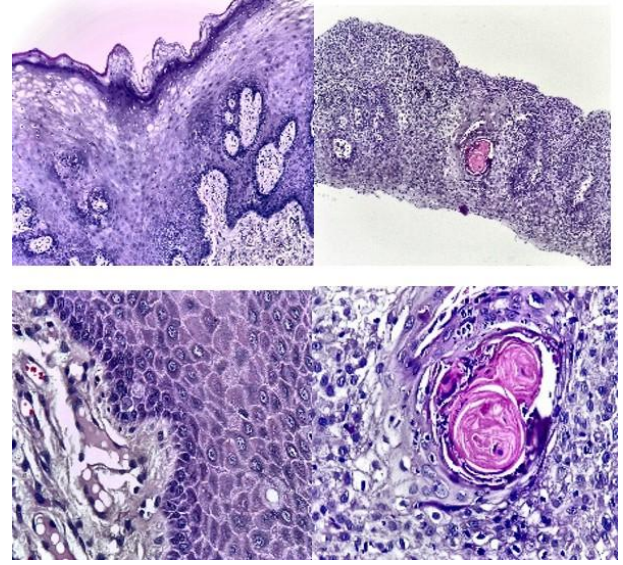


Figure 3: After Preprocessing.

C. General Process Flow

The filtered image is passed into the XSFN-Net Computation block, which consists of a series of four convolutional layers. These convolutional layers are responsible for extracting deep spatial features that capture the intricate patterns and textures unique to cancerous and non-cancerous tissues. To reduce the dimensionality and retain the most prominent features, the model employs an Average Pooling operation with a 3×3 convolution, which also aids in reducing computational complexity and mitigating overfitting.

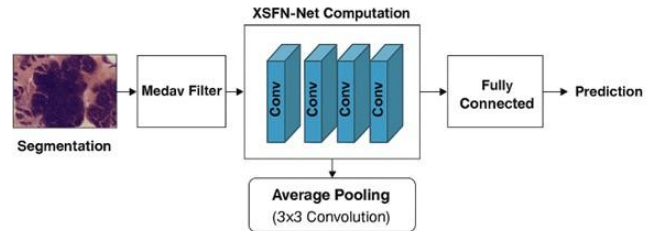


Figure 4: Block Diagram of the Proposed XSFN-NET Architecture

The features are subsequently sent over to a Fully Connected Layer, which transforms the features collected, into a one-dimensional vector for classification. In this stage, we will likely see a dropout regularization layer in order to enhance generalizability. Ultimately, the training model produces a

prediction through the prediction layer, based on the learned features, to determine if the image is Normal or OSCC. This end-to-end approach relies on automatic preprocessing and deep feature extraction to provide a fast and accurate solution for OSCC detection.

D. Model Architecture

Xception Spiking Fractional Neural Network, is the hybrid Deep Learning architecture designed for the accurate classification of the Oral Squamous Cell Carcinoma (OSCC) from histopathological images. This model combines the strengths of three powerful concepts: the Xception deep convolutional network, fractional calculus based feature fusion and deep spiking neural networks (DSNNs).

The XSFN-Net model operates as follows: the Medav Filter is first applied to input images - this is a new preprocessing filter that combines the adaptive median filter and means filter for noise reduction and tissue structures preservation. Subsequently the image is segmented into lesion areas (which has typically achieved using PraNet), and then features are extracted from the image in two streams, and attempted parallelism: handcrafted features (DWT, LVP. and GLRM), and deep features.

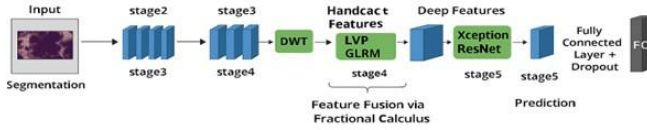


Figure 5: Architecture of the OSCC with Fractional Calculus Feature Fusion

The combination of deep and handcrafted features is done through a fractional calculus-based approach, which gives an edge in comparison to traditional approaches, providing a richer and flexible information integration. This fusion layer effectively combines handcrafted textural patterns with deep spatial features. The fused representation is passed through a fully connected layer with dropout for dimensionality reduction and regularization. Finally, the classifier outputs the prediction as either Normal or OSCC based on the learned patterns.

The calculated architecture is attempting to leverage the best of both domain specific handcrafted features and deep representations, to create a potentially more powerful and effective model for OSCC detection.

The Xception architecture is built upon depthwise separable convolutions, which extract high-level semantic aspects of an

image to allow capturing of granular morphologic distinctions between normal and cancerous tissues. Along with deep learning features, handcrafted features add texture and frequency information, with thier lowerlevel characteristics. The combination of both domains is done through a fraactional calculus-based mechanism, which, in comparison to The generated feature vector is then classified in a biologically-inspired sparse deep neural network model known to replicate the spiking mechanism of biological systems, thus yielding sig- nificant reduction methods, maintains the discriminative power of features more efficiently.

XSFN-Net excels in classifying OSCC with high accuracy, sensitivity, and specificity, making it a powerful tool in early cancer diagnosis. Its combination of handcrafted insights, deep representation learning, and biologically plausible decision-making positions it as an advanced solution in medical im- age analysis, especially in tasks involving fine-grained tissue differentiation such as OSCC detection.

IV. EXPERIMENTAL SETUP

All models were implemented using Python with the PyTorch and TensorFlow frameworks. Training took place in the Google Colab environment, using an NVIDIA Tesla T4 GPU with 16 GB VRAM. We resized all histopathological images to 224×224 pixels and normalized them before passing them to the models.

The models were trained with a batch size of 16 and a maximum of 5 k-folds. The AdamW optimizer was used with an initial learning rate of $1e-4$, and we adopted Cosine Annealing as the learning rate scheduler. To prevent overfitting, we applied early stopping based on validation loss. We also used dropout layers and weight decay ($1e-5$) for regularization.

V. RESULT AND DISCUSSION

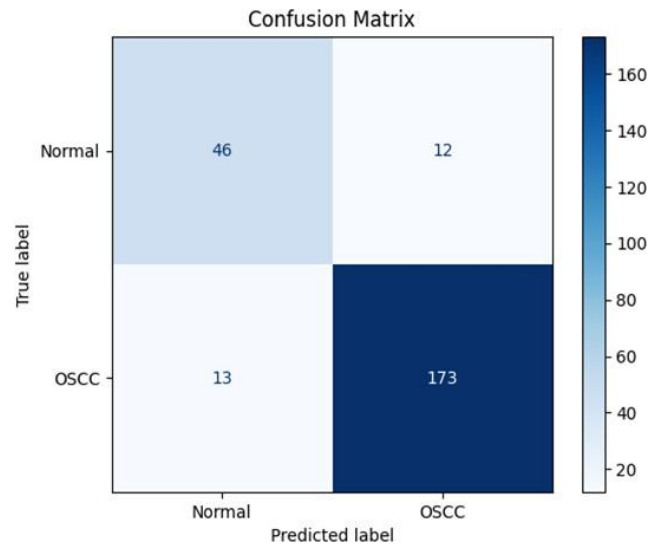


Figure 6: Confusion Matrix

The provided confusion matrix has valuable insights regarding the performance metrics of the OSCC classification model. From the matrix, out of 186 true OSCC cases, 173 forecasted as OSCC, which constitutes true positives, while 13 were underreported as OSCC which makes them false negatives. Among the 58 true normal cases, 46 were accurately identified as normal, while the remaining 12 were misclassified as OSCC.

From this data, OSCC detection precision was estimated at 93.51%. This demonstrates the model's precision when reducing false positives. Sensitivity recall was 93.01% which reflects the model's OSCC detection capability. In addition, the balance of precision and recall, the F1-score, was 93.64% which shows the model consistently identifies cancerous tissues. Overall, it can be said that the model accurately detected OSCC and therefore can be used for clinical and diagnostic purposes.

Table II: Model Performance Comparison Table

| Model | k-folds | Accuracy | Precision | Recall | F1-Score |
|----------|---------|----------|-----------|--------|----------|
| CNN | 5 folds | 76% | 74.5% | 75.2% | 74.8% |
| OralNet | 5 folds | 76.31% | 75.1% | 76.0% | 75.5% |
| XSFN-Net | 5 folds | 91% | 93.51% | 93.01% | 93.64% |

The performance of three different models on the OSCC dataset was evaluated using five-fold cross-validation: the CNN baseline model, OralNet, and the proposed model, XSFN-Net. The baseline CNN model obtained an F1-score of 74.8%, precision of 74.5%, and recall of 75.2%. OralNet is a customized model designed for classifying oral histopathological images. It shows slight improvement over the CNN model, achieving an accuracy of 76.31%, with a precision of 75.1% and a recall of 76.0%.

In all metrics, the proposed model, XSFN-Net, outperformed both baseline models. Primarily, it yielded a precision of 93.51%, recall of 93.01% and F1-score of 93.64%, while obtaining an accuracy rate of 91%.

A. Model Accuracy per Fold

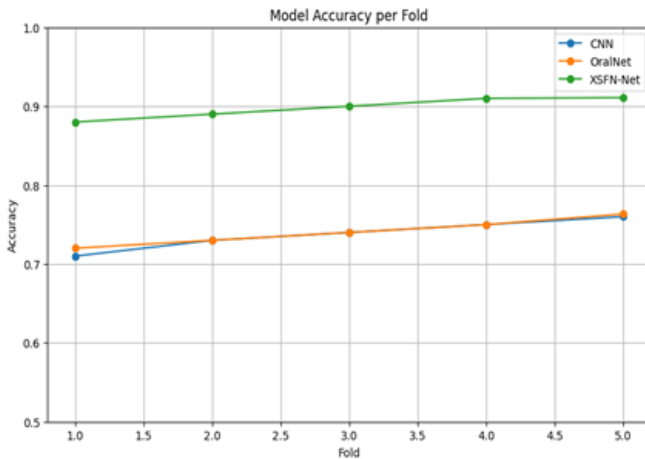


Figure 7: Model Accuracy per Fold

The above Figure shows the accuracy comparison for three models, CNN, OralNet, and XSFN-Net, across five folds using stratified cross-validation on Dataset-1.

The plot shows that XSFN-Net consistently outperforms both CNN and OralNet in all five folds. CNN and OralNet display a gradual increase in accuracy from fold 1 to fold 5, starting around 71% to 72% and rising to about 76%. Their performance remains close and similar throughout the process. In contrast, XSFN-Net begins with a much higher accuracy of about 88% in fold 1 and keeps improving, reaching approximately 91% by fold 5.

This performance gap highlights the effectiveness of the XSFN-Net architecture. It benefits from integrating hand-crafted features, such as LVP, DWT, and GLRM, as well as deep Xception features and classification through a Deep Spiking Neural Network (DSNN). The increasing trend across folds for all models also shows the stability and generalizability of the training setup.

B. Average Accuracy of Models

The Figure shows a comparison of the average classification accuracy achieved by three different models, CNN, OralNet, and the proposed XSFN-Net, on Dataset-1. The average accuracy values were calculated over five stratified cross-validation folds.

As shown in the bar chart, both CNN and OralNet had similar average accuracy scores of about 74%. This suggests comparable performance between these traditional convolutional neural networks. In contrast, the XSFN-Net reached a much higher average accuracy of around 90%, clearly outperforming the other two models.

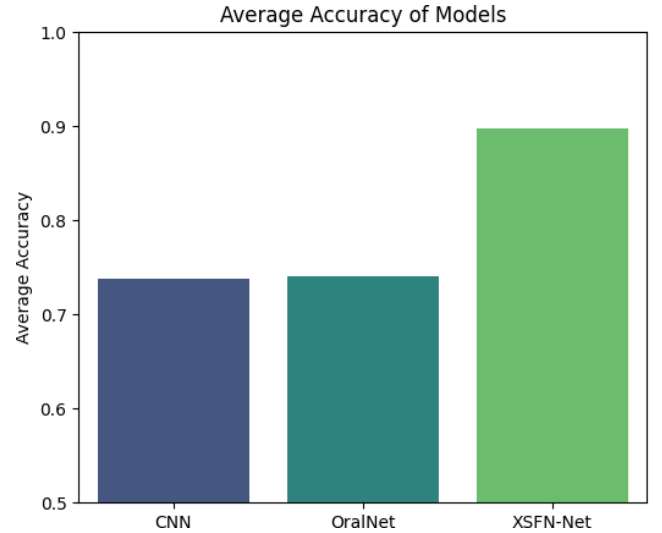


Figure 8: Average Accuracy of Models

The significant performance boost of XSFN-Net comes from its hybrid design, which combines handcrafted features, such as Local Vector Pattern, Discrete Wavelet Transform, and Gray-Level Run Length Matrix, with deep features extracted

using the Xception model. The combination of these features through fractional calculus and the final classification via a Deep Spiking Neural Network (DSNN) adds to its excellent discriminative power. Overall, this figure visually supports the effectiveness of the proposed model compared to traditional deep learning architectures for OSCC classification.

The experimental evaluation took place on Dataset-1 using a 5-fold stratified cross-validation method to check the reliability and generalizability of three different models: CNN, OralNet, and the proposed XSFN-Net. We measured performance with standard classification metrics, including accuracy, precision, recall (sensitivity), and F1-score.

The results show that the XSFN-Net model significantly outperforms both CNN and OralNet across all performance metrics. Specifically, CNN achieved an average accuracy of 76%, with precision at 74.5%, recall at 75.2%, and an F1-score of 74.8%. OralNet had a slight edge over CNN, reaching 76.31% accuracy, 75.1% precision, 76.0% recall, and 75.5% F1-score. In contrast, XSFN-Net reached an accuracy of 91%. Its precision was 93.51%, recall was 93.01%, and the F1-score was 93.64%.

The improvement in performance comes from XSFN-Net's hybrid architecture. This model combines both handcrafted features (LVP, DWT, GLRM) and deep features (Xception). These features are fused using Fractional Calculus and classified with a Deep Spiking Neural Network (DSNN). With this combination, the model can integrate low-level texture granularity and high-level semantic overview, which enhances its differentiating capabilities.

The fold-wise performance, illustrated in Model Accuracy per Fold, XSFN-Net's accuracy stayed consistently high and ranged between 88% to 91%. As for CNN and OralNet, while they did progress, they remained stagnant below 77% accuracy. The Average Accuracy of Models showing average accuracy further highlights that XSFN-Net surpasses the other models in achieving higher classification accuracy without fail.

The combination of crafted and deep features demonstrates the model's histopathological image classification capabilities. Moreover, the model's DSNN component, which draws inspiration from biological systems, enhances the model's accuracy while also incorporating temporal dynamics and energy-efficient computation, which in turn makes the model more efficient for potential real-time use.

VI. CONCLUSION

In this work, we introduced XSFN-Net, a novel hybrid deep learning model designed to accurately classify oral squamous cell carcinoma (OSCC) from histopathological images. In addition to deep features taken from the Xception network, the model is constructed using the handcrafted features LVP, DWT, and GLRM. These features are extracted, fused using fractional calculus, and then classified using a deep spiking neural network (DSNN), which mimics the temporal processing of the brain.

The comprehensive evaluations conducted over five cross-validation folds demonstrated that XSFN-Net performs better

than the standard CNN and OralNet models. The model attained an impressive 91% accuracy alongside significant improvements in precision, recall, and F1-score, confirming that the model is effective in capturing the spatial and frequency-domain features pertinent to OSCC.

The Medav-filtering-augmented, PraNet-segmentation-based, feature-fusion together with DSNN-classification-based pipeline optimization, offers an innovative energy-saving solution inspired by biological processes, while also providing deep learning core enhancements for cancer diagnostics. The solution shows great potential for practical use in clinical settings, such as in the early detection of OSCC in low-resource settings.

Even though XSFN-Net performs well, there are still a number of drawbacks. The study's dataset only includes 1,224 histopathological images, which raises the possibility of overfitting and may limit the model's applicability to larger and more varied clinical cohorts.

Focused on trust-boosting interpretability for the clinical usage of the proposed method, the expansion on multi-class classification deployment for edge devices in real-time is proposed for future work. Multimodal integration could further improve diagnostic accuracy and reliability, and future research should investigate explainable AI (XAI) techniques to further increase transparency in decision-making.

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